

Review on pressure swirl injector in liquid rocket engine

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ABSTRACT

The pressure swirl injector with tangential inlet ports is widely used in liquid rocket engine. Commonly, this type of pressure swirl injector consists of tangential inlet ports, a swirl chamber, a converging spin chamber, and a discharge orifice. The atomization of the liquid propellants includes the formation of liquid film, primary breakup and secondary atomization. And the back pressure and temperature in the combustion chamber could have great influence on the atomization of the injector. What's more, when the combustion instability occurs, the pressure oscillation could further affects the atomization process. This paper reviewed the primary atomization and the performance of the pressure swirl injector, which include the formation of the conical liquid film, the breakup and atomization characteristics of the conical liquid film, the effects of the rocket engine environment, and the response of the injector and atomization on the pressure oscillation.

1. Introduction

Liquid-propellant rocket engines have been used as the primary propulsion systems in most launch vehicles and spacecraft since the late 1920's [1,2], such as the planet landers and low-cost engines [3]. The performance of liquid rocket engine is determined not only by the propellant selection but also by fuel and oxidizer atomization performance [4,5], evaporation and ignition of droplets [6–8]. The atomization performance of propellants is determined by the injector. And there are many types of injector, for example the liquid centered gas-liquid pintle injector [9,10], liquid-liquid pintle injector [11], liquid centered swirl coaxial injector [12,13], etc.

Pressure swirl injectors are extensively used in liquid rocket engines [14], gas turbine engines [15], internal combustion engines [16,17], and many other combustion applications [18]. The pressure swirl injector can be divided into hollow cone injector [19], solid cone injector [20], and spill-return injector [21]. And the swirling motion of liquid can be formed by either tangential inlet ports [22] or a swirler [19]. In liquid rocket engine, the injector configuration should be as simple as possible to ensure the reliability and stability. Thus the pressure swirl injector with tangential inlet ports is widely used in liquid rocket engine. Commonly, this type of pressure swirl injector consists of tangential inlet ports, a swirl chamber, a converging spin chamber, and a discharge orifice [23], as depicted in Fig. 1. And it can be further divided into converge-end swirl injector and open-end swirl injector based on

whether there is a converging spin chamber, as depicted in Fig. 2. The liquid is injected through the tangential ports, forming an air core along the centerline due to high liquid swirl velocity. The liquid flow at the discharge end presumes a hollow conical swirling film. Then the swirling film becomes unstable and breaks up into droplets, as shown in Fig. 1.

The pressure swirl injector is used to atomize the liquid propellants through the formation of liquid film, primary breakup and secondary atomization. Atomization is a process during which the interfacial area of liquid increases gradually because the bulk liquid is transformed into small droplets. So, the evaporation of liquid propellants can be facilitated by atomization significantly. From the energy point of view, atomization is a process during which the potential energy of the supplied liquid finally converts into the needed surface energy, as shown in Fig. 3. Jedelsky and Jicha [21] studied the energy conversion in atomization of a spill-return pressure swirl injector, they found that 58% of the pressure drop converts into the kinetic energy of the rotational motion in the swirl chamber, and the energy loss includes hydraulic loss and friction loss. The kinetic energy of the spray near the injector exit is 32–35% of the inlet energy, which contains the droplet kinetic energy (21–26%) and the entrained air kinetic energy (10–13%). Atomization efficiency is defined as the ratio of surface energy and the inlet energy. It decreases with the increase of pressure drop because the viscous loss increases faster than the surface energy. Commonly, the atomization efficiency is less than 0.3%. For the pressure swirl injector, most of the energy loss occurs in the swirl chamber.

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Nomenclature			
A_t	Total area of the tangential ports, m^2	$\dot{m}_l$	Liquid mass flow rate, kg/s
A	Geometry characteristic constant	μ_l	Dynamic viscosity, $Pa \cdot s$
α	Converge angle of the swirl chamber, deg	ν_l	Kinematic viscosity, m^2/s
β	Spray cone angle, deg	ω	Angular frequency of the surface wave, Hz
β_r	Spray cone angle at a distance L from the injector exit, deg	ω_s	Maximum surface wave growth rate
C_d	Discharge coefficient	ω_i	Surface wave growth rate
d_l	Ligaments diameter, m	PDA	Phase doppler anemometry
d	Droplet diameter produced by the breakup of the ligaments, m	P_c	Chamber pressure, Pa
D_t	Tangential ports diameter, m	Q	Volume flow rate, m^3/s
D_0	Injector diameter, m	R	Radius, m
D_s	Swirl chamber diameter, m	Re	Reynolds number
D_a	Air core diameter, m	R_s	Radius of the swirl chamber, m
D_{at}	Air core diameter at the tangential inlet, m	R_g	Gas constant
ΔP_l	Liquid pressure drop, Pa	R_t	Tangential inlet radius of the injector, m
η	Small disturbance, m	R_0	Orifice radius of the injector, m
η_{bu}	Critical amplitude of the disturbance waves, m	Ro	Rossby number which characterize the ratio of the axial velocity and the rotational velocity
η_0	Initial amplitude of the disturbance waves, m	Re_t	Reynolds number at the tangential inlet
F_s	Centrifugal force, N	Re_{th}	Theoretical Reynolds number of the liquid film at the injector exit
F_{scr1}	The critical centrifugal force for judging air core, N	R_{sw}	Swirling radius of the pressure swirl injector, m
F_{scr2}	The critical centrifugal force for judging stable air core, N	ρ_l	Liquid density, kg/m^3
$GSMD$	Global Sauter Mean Diameter, μm	ρ_g	Gas density, kg/m^3
GLR	Gas liquid ratio	SMD	Sauter Mean Diameter, μm
γ	Opening coefficients	S_n	Swirl number
h	Film thickness, m	σ	Surface tension coefficient
h_0	Film thickness at the injector exit, m	σ_{cr}	Critical pressure ratio
h_t	Film thickness at the inlet of the tangential ports, m	τ_{bu}	Breakup time, s
K	Injector constant	φ	Filling coefficient
K_v	Velocity correction coefficient	ϑ	Angle of the tangential ports, deg
K_s	Wave number correspond to the maximum growth rate	We	Webber number
L_0	Orifice length, m	We_l	Liquid Webber number
L_s	Swirl chamber length, m	W	Tangential velocity, m/s
L_{bu}	Slant breakup length, m	W_t	Velocity of the tangential inlet, m/s
L_v	Vertical breakup length of the conical liquid film, m	X	Dimensionless air core diameter
λ	Wavelength of the surface wave, m		

The operating principle of liquid rocket engine is quite different from other combustion applications, making the operation condition of pressure swirl injector in liquid rocket engine quite different. For example, in automobile engine, the spray is pulse and the pressure drop is really high. And the research focuses on the atomization of biofuel [24–26], secondary injection [27] and the effects of injector geometry [28–30] for energy-saving and emission reduction. However, in liquid rocket engine, the spray is stable, both the pressure drop and the ambient temperature are high. What's more, when throttling process or combustion instability occurs, the pressure drop and chamber pressure will also vary. These characteristics indicate that the liquid propellants could be super critical and the pressure in supply system and combustion chamber could be oscillating.

Although the pressure swirl injector has been reviewed before. For example, Vijay et al. [31] reviewed the injector geometrical parameters, fluid properties and operating conditions' influence over the air core stability, breakup length, spray cone angle and Sauter mean diameter. The effects of the rocket operating environments on the atomization mechanism and spray characteristics of the pressure swirl injector has not been summarized before. In this paper, a more comprehensive review on the pressure swirl injector in liquid rocket engine has been conducted. The internal flow characteristics, conical film formation, primary breakup and spray characteristics has been discussed. Then the effects of rocket operating environments on the spray characteristics have been discussed further. These operating environments include back pressure, super critical injection and pressure oscillation.

2. Discussion

2.1. Formation of conical liquid film

In the pressure swirl injector, the liquid is injected through the tangential ports, forming an air core and an annular liquid film along the centerline due to high liquid swirl velocity. Then, the annular liquid film develops into hollow conical swirling film when it flows out of the injector exit. The formation of conical liquid film is strongly related with the internal flow characteristics which includes the air core formation [31], the boundary layer development [23], the film thickness and the growth of the unstable wave at the interface between the gas and liquid, as shown in Fig. 4.

2.1.1. Air core formation

An air core is formed when the centrifugal force of the swirling flow overcomes the viscous force and a low-pressure area near the injector exit is created by the centrifugal motion of liquid within the swirl chamber [23]. The centrifugal force can be calculated by $dF_s = dm \frac{W^2}{r}$, and is proportional with the square of the tangential velocity W . The tangential velocity at a specific radius position W is proportional with the velocity at the tangential inlet ports W_t for the angular momentum conservation $Wr = W_t R_{sw}$. It means that the centrifugal force increases with the increase of the tangential inlet velocity W_t , the tangential inlet Reynolds

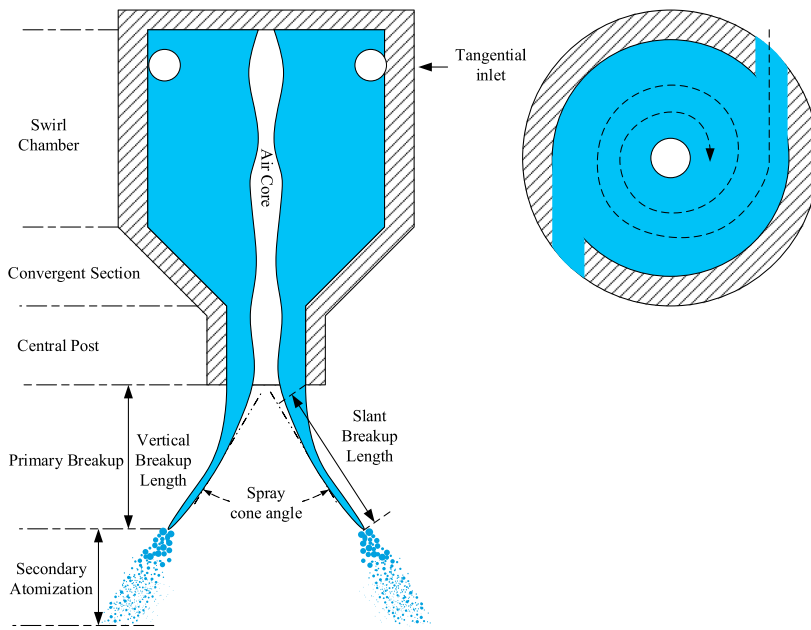


Fig. 1. Schematic of the flow formed by a pressure swirl injector.

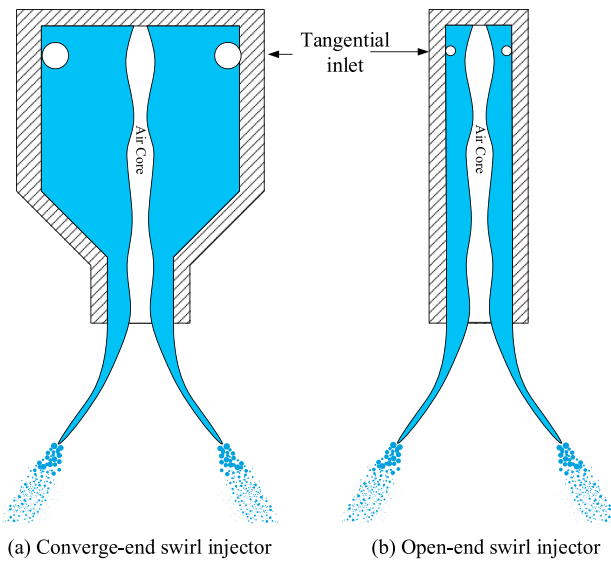


Fig. 2. Schematic of two typical pressure swirl injectors.

number $Re_t = \frac{\rho_l W_t D_t}{\mu_l}$, the pressure drop and the swirl number $S_n = \int V W r^2 dr / \int r_0 V^2 r dr$.

Table 1 shows the air core formation mechanism of pressure swirl injector with different operation conditions and geometrical parameters. Datta and Som [32] found the pressure drop rapid increases when it exceeds a critical value. Moon et al. [30] figured out that a stable air core is formed when the swirl number $S_n > 0.6$. Typical development of air core with the increase of Reynolds number is show in Fig. 5. Researches of Halder et al. [33], Lee et al. [34] and Amini et al. [23] indicate that the flow regime develops from no air core to fully developed stable air core with the increase of liquid Reynolds number, and a transition stage with a developing air core locates between these two regimes. It means that there are two critical centrifugal forces: F_{scr1} and F_{scr2} . When the

centrifugal force is smaller than F_{scr1} , no air core is formed. When the centrifugal force locates between F_{scr1} and F_{scr2} , a developing unstable air core is formed. When the centrifugal force is larger than F_{scr2} , a fully developed stable air core is formed. The fully developed stable air core is cylindrical and bulges in diameter at the entrance of the discharge orifice [33]. The geometrical parameters and liquid viscosity have great influence on the formation of air core. When the ratio of the swirl chamber length to the swirl chamber diameter is large enough and the initial angular momentum is smaller than the needed one for a stable air core, a double-helical unstable air core is formed [35,36], as shown in Fig. 6. Liquid viscosity hinders the formation of air core [37], because the viscosity decreases the velocity of the swirling flow [38].

The diameter of the fully developed air core increases with the increase of liquid Reynolds number and tends to a constant [23,32,33]. And among the geometrical parameters of pressure swirl injector, the injector constant $K = \frac{A_t}{(D_s - D_t)D_0}$ has the greatest influence on the internal flow characteristics. For example, the air core diameter decreases with the increase of injector constant K [23]. That's why the air core diameter increases with the increase of injector diameter D_0 , the decrease of tangential ports diameter D_t and the increase of swirl chamber diameter D_s [33]. Halder [33] obtained the empirical equation of air core diameter from the experimental data:

$$\frac{D_a}{D_0} = 0.338 \left(1 - e^{-1.45 \times 10^{-4} Re} \right) \alpha^{0.073} \left(\frac{D_0}{D_s} \right)^{0.424} \left(\frac{D_t}{D_s} \right)^{-0.732} \left(\frac{L_0}{D_s} \right)^{-0.252} \quad (1)$$

It is clear that besides the injector constant K , the converge angle of swirl chamber α and the orifice length L_0 also have great influence on the air core. The air core diameter D_a increases with the increase of the swirl chamber converge angle and decreases with the increase of orifice length.

2.1.2. Film thickness

The film thickness at the injector exit is an initial parameter of the conical liquid film, thus have great influence on the breakup of conical film. There are three methods to measure the film thickness at the injector exit, as shown in Table 2.

As the radius of the air core plus the film thickness equal to the radius of the injector, the variation of film thickness shows an opposite tendency

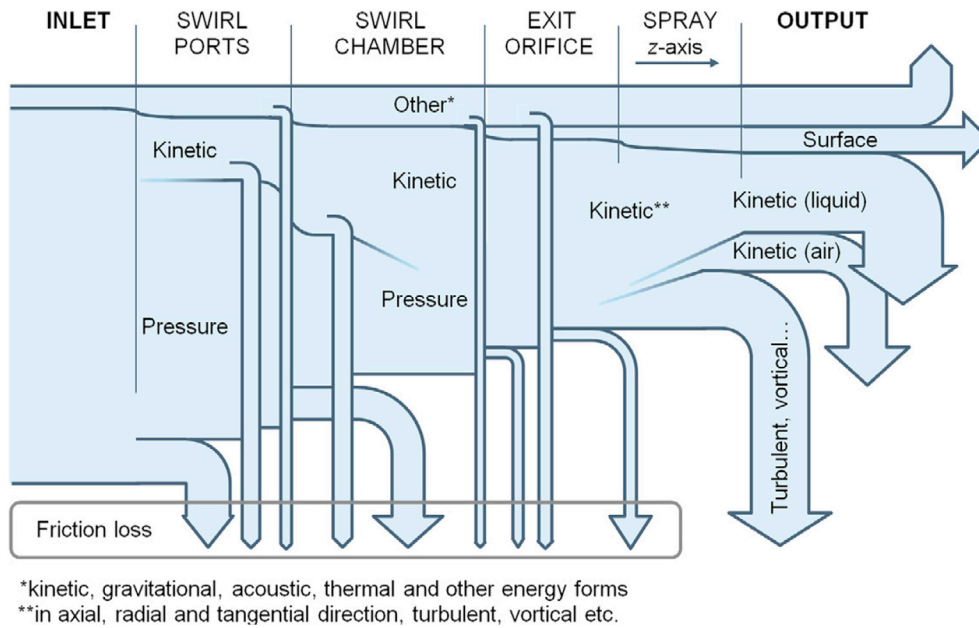


Fig. 3. Sankey diagram for energy balance of the atomization process [21].

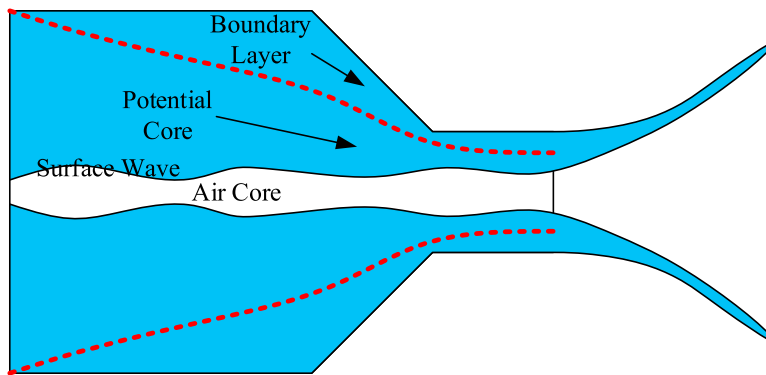


Fig. 4. Schematic of the inner flow of a pressure swirl injector.

with the variation of air core. In the orifice of the injector, the film thickness is about $1000 \mu\text{m}$ and decreases along stream [47]. The film thickness is influenced by both the operation condition and injector geometry significantly. On the point of view of the operation condition, the film thickness decreases with the increase of liquid Reynolds number and pressure drop, and tends to a constant value [23,33,37]. And it increases with the increase of liquid viscosity [37]. On the point of view of the injector geometry, injector constant K has the most significant influence, the length of orifice and the converge angle of the swirl chamber also have great influence. The film thickness increases with the increase of injector constant K [23,32,33,48], increases with the increase of orifice length L_0 and decreases with the increase of swirl chamber converge angle α [23,32,33].

Using theoretical and experimental methods, researchers obtained several empirical equations to predict the film thickness at the injector exit, as shown in Table 3. Where h_0 is the film thickness at the injector exit, μ_l is the liquid viscosity, $\dot{m}_l$ is the liquid mass flow rate, D_0 is the injector diameter, ΔP_l is the pressure drop, ρ_l is the density of liquid and L_0 is the orifice length. These equations indicate that the film thickness h_0 is proportional with the term $\left(\frac{\dot{m}_l \mu_l D_0}{\rho_l \Delta P_l}\right)^{0.25}$, while the coefficient varies between different injectors. Among these empirical equations, the

equation proposed by Kim et al. [35,36] includes the most geometrical parameters of the injector and predict the film thickness well. However, these equations is derived based on the assumption that the azimuthal distribution of the film thickness is uniform and is different with the actual situation. In fact, Inamura et al. [57,58] found that the azimuthal distribution of the film thickness is nonuniform. Moreover, the film thickness fluctuates with peaks and troughs, and the number of peaks is correspond to the number of tangential inlet ports.

2.1.3. Boundary layer

Once the fully developed stable air core is formed, the viscous internal flow of the pressure swirl injector can be divided into boundary layer and potential core [23], as shown in Fig. 4. The boundary layer contains axial flow, while whether the potential core contains axial flow has not been settled. Amini [23] has compared three theoretical models: inviscid flow, viscous flow without the axial flow in the potential core, and viscous flow with the axial flow in the potential core, and found the viscous flow with the axial flow in the potential core agrees better with the simulation results. The dimensionless thickness of the boundary layer obtained by Amini with the axial flow in the potential core considered is shown in Fig. 7. It is clear that the thickness of the boundary layer in the swirl chamber and the orifice of the injector increases gradually along the

Table 1
Regimes of air core.

Author	Type of data and Reference	Regimes	Constraints	Fluid medium
Datta and Som [32]	Theoretical	1. Increasing air core radius 2. Stable air core	The pressure drop ΔP_t keeps constant The pressure drop ΔP_t rapid increases when it exceeds a critical value	
Moon et al. [30]	Experimental	1. Fluctuating air core	$S_n < 0.6$	Gasoline
	Numerical	2. Stable air core	$S_n > 0.6$	
Halder et al. [33]	Experimental	1. No air core	$Re_t < 800$ $0.25 < \frac{D_t}{D_0} < 0.38$	Water
		2. Transition stage with a developing air core	$800 < Re_t < 2400$ $0.25 < \frac{D_t}{D_0} < 0.38$	
		3. Fully developed stable air core	$Re_t > 2400$ $0.25 < \frac{D_t}{D_0} < 0.38$	
Lee et al. [34]	Experimental	1. No air core	$Re_{th} = \frac{\rho_l V_n D_0}{\mu_l} < 2550$ $\Delta P_t < 0.5 \text{ MPa}$	Diesel
		2. Transition stage with a developing air core	$2550 < Re_{th} < 3450$ $0.5 \text{ MPa} < \Delta P_t < 0.9 \text{ MPa}$	bunker-A
		3. Fully developed stable air core	$Re_{th} > 3450$ $\Delta P_t > 0.9 \text{ MPa}$	
Amini et al. [23]	Theoretical	1. No air core	$Re_t < 80$	Water
	Experimental	2. Transition stage with a developing air core	$80 < Re_t < 144$	
		3. Fully developed stable air core	$Re_t > 186$	
Kim et al. [35,36]	Numerical	1. Cylindrical stable air core	$0.7 < \frac{L}{D_0} < 1.06$	Water
	Experimental	2. Double-helical unstable air core	$1.27 < \frac{L}{D_0} < 3.06$	

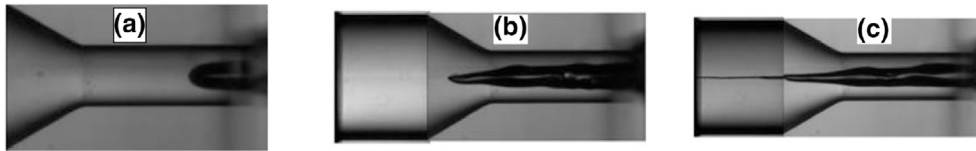


Fig. 5. Air core develops with the increase of Reynolds number: (a) $Re = 101$, (b) $Re = 144$, (c) $Re = 186$ [23].

stream, while that in the convergent section decreases along the stream. Dumouchel et al. [61] found that the boundary layer inside the injector are both functions of the injector geometrical parameters and the pressure drop. When the pressure drop is small, the thickness of the boundary layer is in the same order of magnitude with the film thickness. And the influence of boundary layer on the flow decreases with the increase of tangential inlet ports diameter D_t , injector diameter D_0 , and pressure drop.

2.1.4. Discharge coefficient

The discharge coefficient directly reflects the discharge characteristics of pressure swirl injector, and is defined as the ratio of the actual mass flow rate to the theoretical mass flow rate: $C_d = \dot{m}_l / \left(\frac{\pi D_0^2}{4} \sqrt{2 \rho_l \Delta P_t} \right)$.

For inviscid flow, the discharge coefficient is only related with the geometrical parameters. Giffen and Abramovich [62] derived the equation of discharge coefficient based on the inviscid assumption:

$$C_d = \sqrt{\frac{(1-X)^3}{1+X}} = \sqrt{\frac{\varphi^3}{2-\varphi}} \quad (2)$$

where X is the dimensionless air core area, $\varphi = 1 - X$ is the filling coefficient and can be calculated with the geometry characteristics constant A by

$$A = \frac{(1-\varphi)\sqrt{2}}{\varphi\sqrt{\varphi}} \quad (3)$$

the geometry characteristics constant A is the most important parameter of pressure swirl injector, and is defined as:

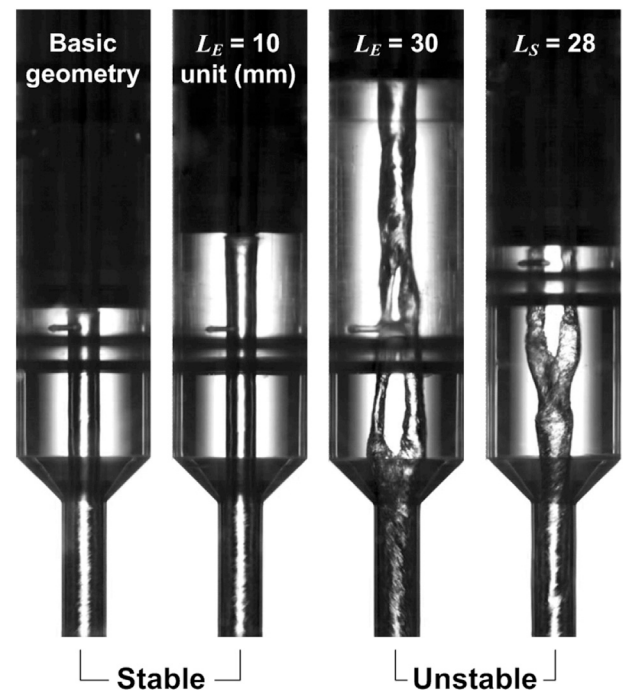


Fig. 6. Typical inner flow of pressure swirl injector [35,36].

Table 2

Measurement method for the film thickness at the injector exit.

Method	Principle	Author	Advantages and Disadvantages
Electrical conductance method	1. Two electrodes are fixed inside the orifice; 2. The voltage for the liquid film thickness was measured; 3. Calculate the film thickness with the calibrated relation between the film thickness and the voltage.	Fu et al. [39–41] Khil et al. [42–44] Chung et al. [45] Kim et al. [35,36]	Advantages: 1. High accuracy of measurement, transient measurement. Disadvantages: 1. The relation between the film thickness and the voltage should be calibrated before, and demand high accuracy of the calibration. So, the standard posts used for calibration should be concentric with the injector, because eccentric setup have great influence on the accuracy [46]. 2. The frequency of the film thickness variation is limited because the distance between the electrodes is finitude. If the distance between the electrodes is much larger than the wavelength of surface wave, only the average film thickness can be measured.
Optical measurement	Transparent acrylic injector: 1. Injector or the orifice of the injector is made by transparent acrylic material; 2. High resolution instantaneous images of the internal flow are captured with high speed camera; 3. The film thickness is directly measured and calibrated.	Feikema [47] Jeng et al. [48] Yao and Fang [37] Kenny et al. [49,50] Moon et al. [51]	Advantages: 1. Transient measurement, can measure the film thickness varying with high frequency. Disadvantages: 1. Three-dimensional overlay effects the visual demarcation and the threshold gray value chosen also causes deviation. 2. Refraction occurs among transparent nozzle, fuel and air, direct measured film thickness is not accurate and optical calibration is needed.
Optical measurement	PLIF: 1. The orifice of the injector is made by transparent acrylic material; 2. Seed the flow with a fluorescent dye; 3. Raw PLIF images are obtained by the camera when the fluorescent dye is illuminated by a laser sheet; 4. Optical calibration is conducted with the raw PLIF image and the film thickness is obtained.	Zadrazil et al. [52] Haber et al. [53] Schubring et al. [54,55]	Advantages: 1. Without the three-dimensional overlay effect, thus this method has high accuracy and can measure transient film thickness; 2. Surface wave structure on the gas-liquid interface can be captured clearly. Disadvantages: 1. The experimental facility is complex; 2. The threshold value chosen causes deviation; 3. Refraction occurs among transparent nozzle, fuel and air, direct measured film thickness is not accurate and optical calibration is needed.
	Image processing: 1. Capture the raw images with long exposures and deconvolute the digital images with an Abel transform technique; 2. Choose a threshold value and obtain the film thickness.	Eberhart et al. [56]	Advantages: 1. Film thickness downstream the injector exit can be measured. Disadvantages: 1. Only the average film thickness is measured; 2. The choose of threshold value causes deviation.
Contact needle probe method	1. Move the contact needle probe to touch the liquid film and record the position when the needle touches the film; 2. Calculate the film thickness with the position of the film surface.	Inamura et al. [57,58]	Disadvantages: 1. The needle probe would influence the internal flow; 2. Only the average film thickness is measured.

$$A = \frac{(D_s - D_t)D_0}{4nR_t^2} = \frac{(Q/n\pi R_t^2)}{Q/\pi R_0^2} \frac{D_s - D_t}{D_0} = \frac{V_{\text{tangential}}}{V_{\text{axial}}} C_{\text{open}} \quad (4)$$

For viscous flow, the discharge coefficient of the pressure swirl injector with tangential ports decreases with the increase of pressure drop, and tends to a constant [63]. The reason is that during the formation of air core, the diameter of the air core increases gradually, which decreases the filling coefficient. Lee et al. [34] found that the discharge coefficient decreases with the increase of Reynolds number Re at the inlet of the tangential ports. And when the flow develops from the unstable stage without air core to the stable stage with air core, the discharge coefficient decreases the fastest, as shown in Fig. 8. What's interesting is that the discharge coefficient of a pressure swirl injector with a swirler has the inverse trend with the pressure swirl injector with tangential ports. Namely, the discharge coefficient of a pressure swirl injector with a swirler increases with the increase of pressure drop [64]. It is because that the larger the pressure drop, the larger the proportion of the axial

momentum and the larger the discharge coefficient.

When the discharge coefficient becomes constant, it is no longer influenced by the operation condition and only influenced by the injector geometry. The most important geometrical parameters of pressure swirl injector includes the geometry characteristics constant A and the injector constant K . And the injector constant K is defined as:

$$K = \frac{A_t}{(D_s - D_t)D_0} \quad (5)$$

While the geometry characteristics constant A and the injector constant K satisfies $A = \frac{\pi}{4K}$, the variation trends of discharge coefficient with the increase of these two parameters are on the contrary. Namely, the discharge coefficient decreases with the increase of the geometry characteristics constant A [65], while it increases with the increase of the injector constant K [48]. Giffen et al. [15], Rizk and Lefebvre [59], Abramovich [62], Fu [66], Liu [67], Jones [68] and Benjamin et al. [69, 70] proposed the empirical equations of the discharge coefficient from

Table 3

Empirical equations of the film thickness at the injector exit.

Author	Equation	Method to obtain	Injector type
Rizk and Lefebvre [59]	$h_e = \left[\frac{1560\mu_t \dot{m}_t}{D_t \Delta P_t} \frac{1+\phi}{(1-\phi)^2} \right]^{0.5}, \phi = \left(1 - \frac{2h_e}{D_t} \right)^2$	Theoretical derivation	Converge-end
Rizk and Lefebvre [59]	$h_e = 3.66 \left(\frac{D_t \dot{m}_t \mu_t}{\rho_t \Delta P_t} \right)^{0.25}$	Theoretical derivation	Converge-end
Suyari and Lefebvre [60]	$h_e = 2.7 \left(\frac{D_t \dot{m}_t \mu_t}{\rho_t \Delta P_t} \right)^{0.25}$	Electrical conductance method	Open-end
Fu et al. [40]	$h_e = 3.1 \left(\frac{D_t \dot{m}_t \mu_t}{\rho_t \Delta P_t} \right)^{0.25}$	Electrical conductance method	Open-end
Kim et al. [35,36]	$h_e = 1.44 D_t \left(\frac{\dot{m}_t \mu_t}{\rho_t \Delta P_t D_t^2} \right)^{0.25} \left(\frac{D_t}{D_s} \right)^{0.6}$	Electrical conductance method	Open-end

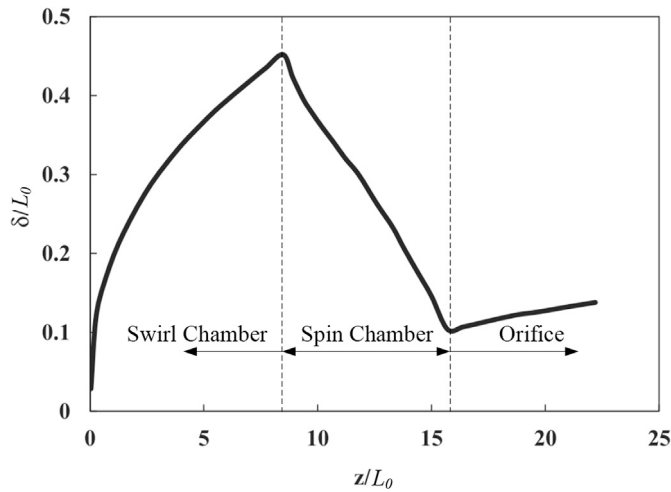


Fig. 7. Dimensionless thickness of the boundary layer in the injector [23].

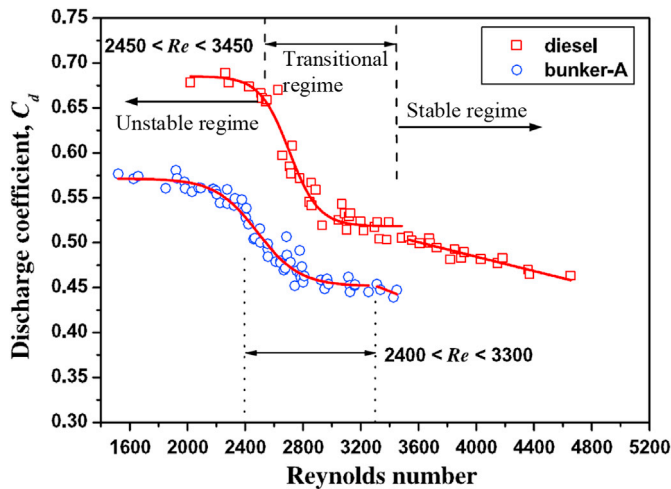


Fig. 8. Discharge coefficient variation with different Reynolds number at the tangential inlet port [34].

experimental data, and are listed in Table 4. It is clear that these equations all predict the variation trend of the discharge coefficient well.

Besides with the geometry characteristics constant A and the injector constant K , other important geometrical parameters includes the injector diameter D_0 , the orifice length L_0 , the swirl chamber length L_s , the

converge angle of swirl chamber α , the area of tangential ports A_t , the opening coefficient $\gamma = \frac{D_t - D_0}{D_0}$, the inlet port angle and so on. The variation pattern of the discharge coefficient with these parameters are as follow:

1. The discharge coefficient decreases with the increase of injector diameter D_0 [32,72], because the increase of injector diameter D_0 will decrease the injector constant K and increase the geometry characteristics constant A , which finally promote the swirling flow and decrease the filling coefficient;
2. The discharge coefficient increases with the increase of the area of tangential ports A_t [32,73], because the swirling flow is weakened by the increase of the tangential ports area, which increases the filling coefficient;
3. The discharge coefficient decreases with the increase of the inlet port angle [15,74], because the increase of the inlet port angle can increase the tangential velocity and decrease the axial velocity, namely, the swirling flow is enhanced;
4. The discharge coefficient decreases with the increase of the opening coefficients γ [75];
5. A lot of research found the discharge coefficient decreases with the increase of the orifice length L_0 [67,68,75], because the longer the orifice length, the larger the friction loss. However, the equation of Benjamin et al. [69,70] indicates that the discharge coefficient increases with the increase of the orifice length. The reason may be that the range of the length to diameter ratio is too small (1.4~2.28);
6. A lot of research found the discharge coefficient increases with the increase of the swirl chamber converge angle α [15,23,74], because the larger the converge angle, the shorter the converge section and the smaller the friction loss. Only the results of Datta and Som [32] shows that the discharge coefficient decrease with the converge angle α ;
7. The effects of the swirl chamber length L_s is not clear, Yule and Widger [72] found that increase the swirl chamber length increases the discharge coefficient. However, Sakman et al. [75] found that increase the length to diameter ratio of the swirl chamber increases the discharge coefficient initially and then decreases;
8. The trumpet at the injector exit has little influence on the discharge coefficient [15,74].

In conclusion, the orifice length L_0 , swirl chamber length L_s , swirl chamber converge angle α and the trumpet have little influence on the discharge coefficient. Thus the variation trends can be easily affected by other factors. That's why the contradictions exist in literature. In injector design, it is acceptable to consider the geometry characteristics constant A and the injector constant K only, because the influence of other important parameters (injector diameter D_0 , area of tangential inlet ports A_t and opening coefficient γ) are included in these two parameters.

Table 4
Empirical equations of the discharge coefficient.

Author	Equation	Method	Injector type
Giffen et al. [15]	$C_d = 1.17 \sqrt{\frac{(1-X)^3}{1+X}}$	Experimental	Converge-end
Rizk and Lefebvre [59]	$C_d = 0.35 \left(\frac{A_t}{D_s D_0} \right)^{0.5} \left(\frac{D_s}{D_0} \right)^{0.25}$	Experimental	Converge-end
Abramovich [62]	$C_d = \frac{0.432}{A^{0.24}}$	Experimental	Converge-end
Fu [66,71]	$C_d = 0.19 \left(\frac{A_t}{D_0} \right)^{0.65} \gamma^{-2.13}, C_d = \frac{0.4354}{A^{0.877}}$	Experimental	Open-end
Hong et al. [62]	$C_d = 0.44 \left(\frac{A_t}{D_0} \right)^{0.84} \gamma^{-0.52}, \gamma = \frac{D_t - D_0}{D_0} < 2.3$	Experimental	Converge-end
Liu [67]	$C_d = 0.721 \left(\frac{1}{A} \right)^{0.416} \left(\frac{L_0}{D_0} \right)^{-0.0558} \left(\frac{D_s}{D_0} \right)^{0.147}$	Experimental	Converge-end
Jones [68]	$C_d = 0.45 \left(\frac{D_0 \rho_l V_0}{\mu_l} \right)^{-0.02} \left(\frac{L_0}{D_0} \right)^{-0.03} \left(\frac{L_s}{D_s} \right)^{0.05} \times \left(\frac{A_t}{D_s D_0} \right)^{0.52} \left(\frac{D_s}{D_0} \right)^{0.23}$	Experimental	Converge-end
Benjamin et al. [69,70]	$C_d = 0.466 \left(\frac{D_0 \rho_l V_0}{\mu_l} \right)^{-0.027} \left(\frac{L_0}{D_0} \right)^{0.229} \left(\frac{L_s}{D_s} \right)^{0.091} \times \left(\frac{A_t}{D_s D_0} \right)^{0.517} \left(\frac{D_s}{D_0} \right)^{0.187}$	Experimental	Converge-end

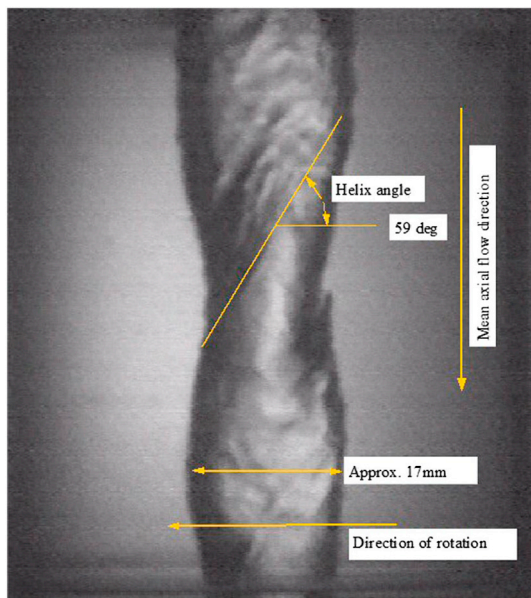


Fig. 9. Air core in the swirl chamber. (Two tangential ports) [78].

2.1.5. Surface wave

In fact, the air core is not cylindrical but spiral, as shown in Fig. 9. Wang et al. [76] and Huo et al. [77] simulated the internal flow of a pressure swirl injector at supercritical conditions, and found the temperature and density iso-surfaces present a spiral structure. Chinn et al. [78] found a helical air core occurs in the injector, and the number of the striation is related with the number of tangential ports. Inamura et al. [57,58] measured the film thickness with the contact needle probe method, and found three film thickness peaks corresponding to the number of the tangential ports. The spiral air core is formed by two reasons. The first one is the nonuniform azimuthal distribution of the tangential velocity, because the larger the tangential velocity, the larger the centrifugal force, and then the larger the air core diameter. As the nonuniform azimuthal distribution of the tangential velocity is produced by the azimuthal distribution of tangential ports, the number of the striation equals the number of tangential ports. The second one is the

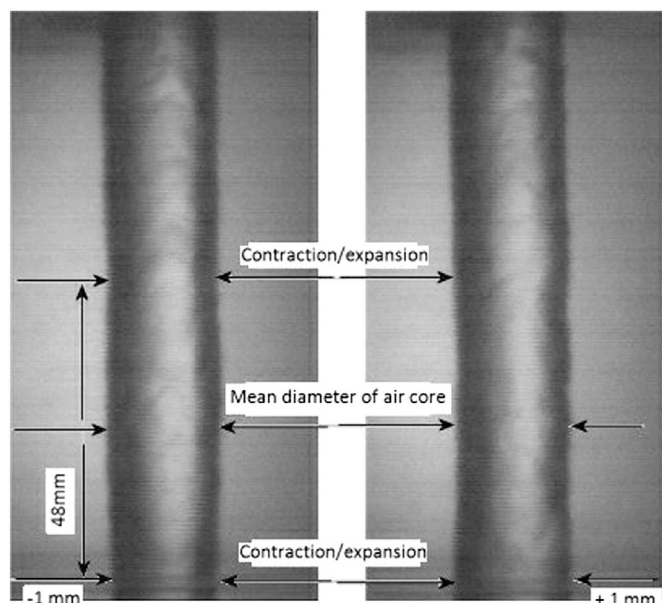


Fig. 10. Stationary pulsating waves in the pressure injector [78].

axial delay of the nonuniform azimuthal distribution of the tangential velocity, because the liquid flow towards the exit with a axial velocity. If the azimuthal distribution of the tangential velocity is uniform, the air core should be cylindrical.

Besides with the spiral air core, stationary pulsating waves also exist on the gas liquid interface in the pressure swirl injector, as shown in Fig. 10. Normally, small disturbances with different frequencies on the gas liquid interface will grow, and the wave with the largest growth rate will dominate the interface. Chinn et al. [78] found the stationary pulsating waves exist on the gas liquid interface. Richardson [79] conducted a linear stability analysis on a confined swirling annular liquid film, and found that the swirling and surface tension are stabilizing forces while gas liquid density ratio is destabilizing force. When the Rossby Number Ro (the ratio of axial liquid velocity to circumferential velocity) is larger than one, the larger the Rossby Number, more unstable the film is. Fu et al. [80] investigated the linear temporal instability of a confined swirling annular liquid film. And found that the injector diameter, Rossby number, and liquid Weber number destabilize the liquid film, while a larger liquid-to-gas density ratio and velocity ratio stabilize the liquid film. Fu [66] further figured out that the film would develop from convectively unstable to absolutely unstable when decrease the liquid-to-gas density ratio, velocity ratio, and liquid Weber number or increase the non-dimensional film thickness and Rossby number Ro .

2.2. Breakup of conical liquid film

The breakup of conical liquid film, also called as the primary breakup, is due to the growth of the unstable wave at the interface between the gas and the liquid film [81]. And the surface waves have already been observed, as shown in Fig. 11. However, the spray pattern and breakup mechanism is much more complex in most cases, because the flow pattern is much more complex than a laminar conical liquid film flow with the influence of the injector geometry and the turbulivity of the flow. The most important parameters of primary breakup are the spray cone angle and the breakup length. The spray cone angle characterize the space distribution range of the liquid film, while the breakup length characterize the breakup position of the liquid film.

2.2.1. Surface wave

There are two independent modes of unstable waves exist on the liquid-gas interface: sinuous wave and varicose wave. While the distortion on two liquid-gas interfaces is same-phase, it is called the sinuous wave (Fig. 12a). While the phase contrast of distortion on two interfaces is 180° , it is called the varicose wave (Fig. 12b). Linear stability analysis has long been used to analyze the development of the unstable waves on the liquid-gas interface [82]. A lot of stability analysis have been focused on round jet, planar liquid film and annular liquid film, while the stability analysis on conical liquid film was limited. Yue and Yang [83] derived the dispersion equation of a conical liquid film and solved the equation numerically. In his derivation, the spray cone angle, the injector diameter, the film thickness at the injector exit and the film thinning along stream has been considered. Wang [84] also derived the dispersion equation of a conical liquid film, but the film thinning is not considered. Fu [66,81] obtained the same dispersion equation of conical liquid film with Yue and Yang [83], and solved the equation numerically. His research indicates that the conical liquid film is dominated by sinuous wave, and the maximum growth rate and dominant wave number increase with the increase of pressure drop, which in turn decrease the breakup time and breakup length. Hosseinalipour et al. [85] further derived the dispersion equation of a conical liquid film with coaxial gas flow, and found that the liquid swirling and gas liquid density ratio are the destabilize force. What's more, a larger spray cone angle destabilizes the conical liquid film and decreases the breakup length. However, the film thinning along stream is also not considered.

When the spray cone angle decreases to zero, the conical liquid film can be looked as an annular liquid film. So the annular liquid film is a

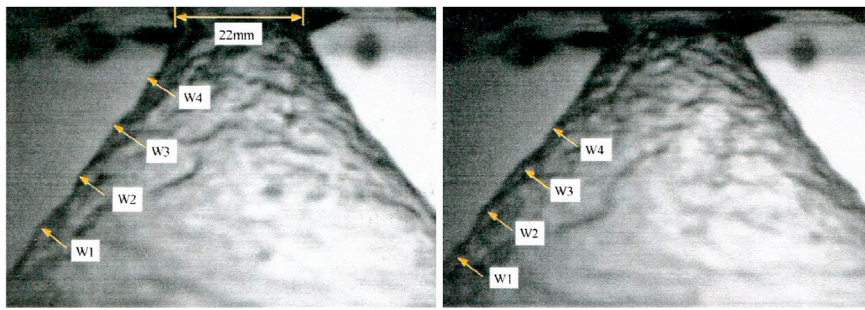


Fig. 11. Surface waves on the conical liquid film [78].

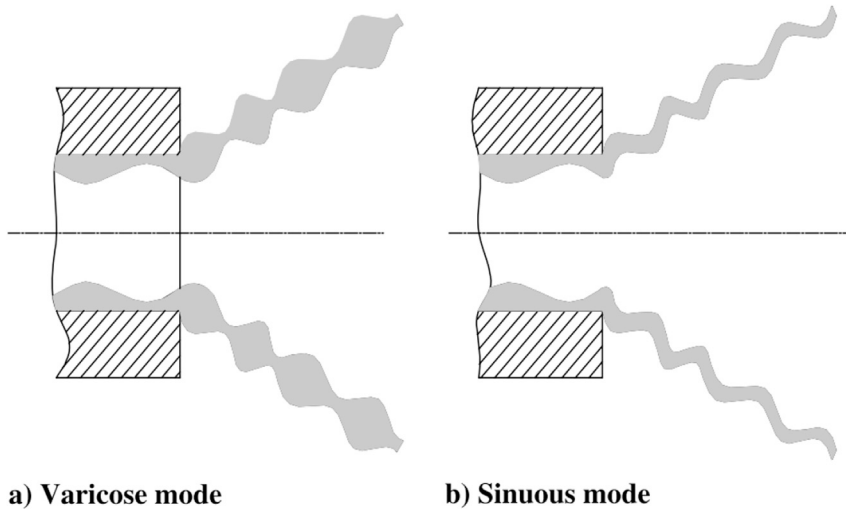


Fig. 12. Wave modes on the conical liquid film [81].

special conical liquid film and the linear stability analysis on annular liquid film could help understanding the growth of surface waves. Chauhan et al. [86] found that the annular liquid film is absolutely unstable for small velocities and convectively unstable for larger velocities. And with the increase of liquid velocity, temporal and spatial instability analysis tend to coincide. Yan et al. [87] and Lin et al. [88] also found that the liquid velocity destabilize the surface and promote the breakup of liquid film. In some cases, the film is swirl. Liao et al. [89] found the liquid swirl not only destabilize the surface but also shifts the dominant wave from axisymmetric mode to helical mode. Lin et al. [88] also found that the liquid swirl could greatly promote the destabilization and disintegration of liquid film, and the dominant mode of annular liquid film is nonaxisymmetric mode. Ibrahim and Jog [90], further figured out that the liquid swirl could destabilize the outer surface and stabilize the inner surface of the annular liquid film.

Besides the liquid flow, the radius of the annular film, the liquid viscosity, gas liquid density ratio and the surface tension all have great influence on the instability of surface waves. Crapper et al. [91] and Shen and Li [92] found that both the growth rate of axisymmetric and non-axisymmetric disturbances rapid increase with the decrease of the radius of the annular liquid film. Yan and Xie [93] also found that the smaller the liquid film radius to film thickness ratio, the more unstable the annular liquid film. Jeandel and Dumouchel [94], Shen and Li [95] and Herrero et al. [96] all found that the liquid viscosity can stabilize the liquid film and suppress the growth of surface waves. Shen and Li [95], Lin et al. [88], Yan and Xie [93], Ibrahim and Jog [90], and Shen and Li [92] found that the gas liquid density ratio can destabilize the liquid film. As for the effects of surface tension, Shen and Li [95] figured out that the surface tension is the stabilize force. However, Shen and Li [92] found that there exists a critical Webber number. The surface tension destabilize the gas liquid interface when the Webber number is smaller than the

critical Webber number, while it stabilize the gas liquid interface when the Webber number is larger than the critical one. It means that the aerodynamic force determines whether the surface tension destabilize or stabilize the gas liquid interface.

2.2.2. Spray pattern and breakup mechanism

In most cases, the spray is not regular cone because the spray pattern is influenced by the centrifugal force and surface tension. And the breakup of liquid film is not solely due to the growth of the unstable wave at the gas liquid interface, because the flow at the injector exit could be turbulent. In fact, the breakup mechanism and the spray pattern are related with each other. Typical spray patterns and breakup processes of a conical liquid film are shown in Fig. 13. Prakash et al. [97] found five spray patterns: dribbling stage, distorted pencil stage, onion stage, tulip stage and fully developed stage. And these spray patterns occur in turn with the increase of liquid Webber number. Ghorbanian et al. [19] observed another wavy stage locates between the tulip stage and fully developed cone stage. Ramamurthi et al. [98] further figured out that the spray will transfer from tulip to cone shape when the liquid Webber number exceeds about 150. Then, Dumouchel [99] found that there exist another perforated cone spray. Kang et al. [100] further figured out that the perforated cone stage followed the wavy stage with the increase of pressure drop, and the spray finally develops into turbulent cone spray (also called as fully developed cone spray).

Above all, there are seven spray patterns: dribbling, distorted pencil, onion, tulip, wavy cone, perforated cone and fully developed cone. These spray patterns occur in turn with the increase of pressure drop, and only the last five spray patterns have cone shape. Furthermore, if the liquid is gelled, the spray patterns transfer to swirling jet, twisted ribbon, fluid web and fully developed hollow cone [65].

The breakup mechanism of a conical liquid film mainly includes

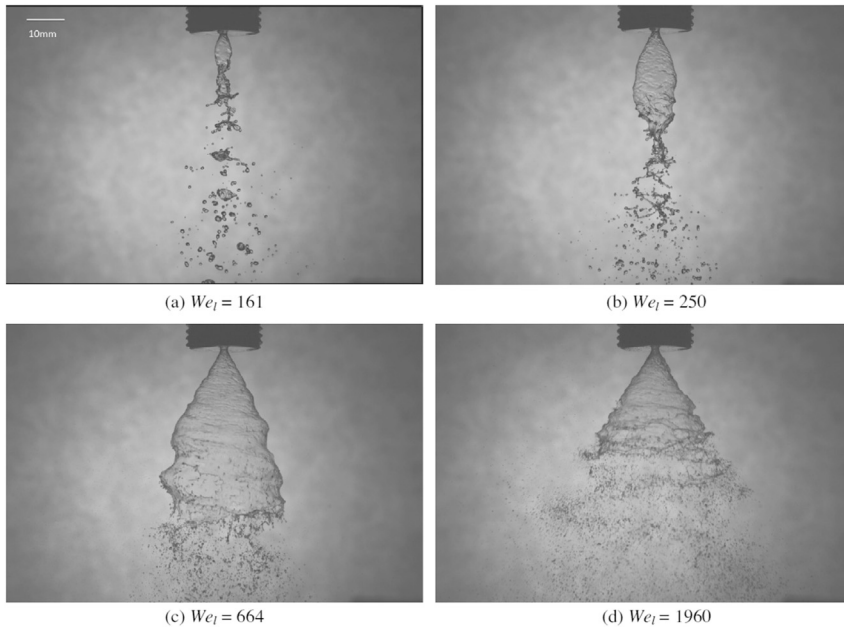


Fig. 13. Typical spray patterns and breakup processes of a pressure swirl injector [97].

surface wave instability, perforation and turbulence. The surface wave instability regime think that the conical liquid film is disintegrated by the growth of surface wave, the perforation regime think that the liquid film is disintegrated by the perforation, while the turbulence regime think that droplets are produced from the liquid film by its own turbulence. Yue et al. [101] found that the conical liquid film breakup at half wavelength. Both short and long wavelength waves exist on the liquid film and influence the breakup simultaneously. Moon et al. [102] found that long-wave regime dominant the breakup under atmospheric pressure while short-wave regime play a dominant role under high ambient pressure. Santolaya et al. [103] analyzed the surface wave instability regime and perforation regime, and figured out that the surface wave instability regime significantly increase the atomization performance and radial distribution of droplets. However, these research mentioned before did not connect the breakup regimes with the spray patterns, but in fact, they are interrelated with each other. For example, the breakup regime of a wavy cone film is the surface wave instability while that of a perforated cone film is perforation. Ramamurthi et al. [98] compared the breakup process of a tulip film and a wavy cone film, and found that the surface wave grow rapidly when the spray pattern changes from tulip to wavy cone. The disintegration of wavy cone film is caused by the growth of surface wave and is different from the tulip film. Kang et al. [100] further figured out that when the spray has an onion shape, the growth of 'Impact Wave' produced by the impingement will finally pinch off the collapsed liquid jet. When the spray has a tulip shape or fully developed wavy cone shape, the breakup regime is surface wave instability. When the spray has a perforated cone shape, the breakup regime is mainly perforation and surface wave instability as a supplement. When the spray has a turbulent cone shape, the breakup regime is mainly turbulence, with perforation and surface wave instability as a supplement.

Besides with these breakup regimes, Garcia et al. [104] and Le et al. [105] investigated the breakup of liquid film from the fractal view, and found that the liquid film is a fractal object and the breakup of liquid film is a fractal process.

2.2.3. Spray cone angle

The spray cone angle is one of the most important parameters of spray characteristics. It controls the space distribution of the spray and has significant influence on the primary breakup of conical liquid film. For inviscid flow, Abramovich [106] derived the equation of the spray cone angle:

$$\tan \beta = \frac{2\sqrt{2}(1-\varphi)}{\sqrt{\varphi}(1+\sqrt{1-\varphi})} \quad (6)$$

where β is the spray half angle. Giffen and Muraszew [107] derived another equations:

$$\begin{aligned} \tan \beta &= \frac{\pi(1-\sqrt{X})}{2K} \\ \sin \beta &= \frac{\pi C_d}{2K(1+\sqrt{X})} \end{aligned} \quad (7)$$

Xue et al. [15] introduced the effects of the tangential ports angle ϑ , and obtained the calculation equation, as listed in Table 5. Rizk and Lefebvre [59] introduced the correction factor of velocity K_v and derived the following equation:

$$\cos \beta = \frac{C_d}{K_v(1-X)} = \sqrt{\frac{1-X}{1+X}} \quad (8)$$

All the equations above are based on the inviscid flow in which the air core diameter at the injector exit equals to the air core diameter in the injector. However, the air core diameter at the injector exit is larger than that in the injector, because the flow conservation is satisfied when across the injector exit, and the liquid film is accelerated by the dynamic pressure that transferred from the centrifugal overpressure. Orzechowski (Chinn [107] and Moon et al. [102]) considered this effect and derived an equation of the spray cone angle:

$$\tan \beta = \frac{2C_d A}{\sqrt{(1+S_b)^2 - 4C_d^2 A^2}} \quad (9)$$

where $S_b = \frac{R_a}{R_0}$ = air core radius at the injector exit/injector radius.

Besides the theoretical derivation, there are several methods to estimate the spray cone angle by image processing in experiments. Usually straight lines are fitted through parts of the spray edge and the angle between them defines the spray cone angle [110], as shown in Fig. 14. Pastor et al. [111] defined the angle of two straight lines which are fitted to the first 60% of the spray closest to the nozzle as the spray cone angle. Kang et al. [112] defined the spray cone angle as the angle of an imaginary spray cone that possesses the same magnitude of cross-sectional area to the spray area when the height of spray cone is spray

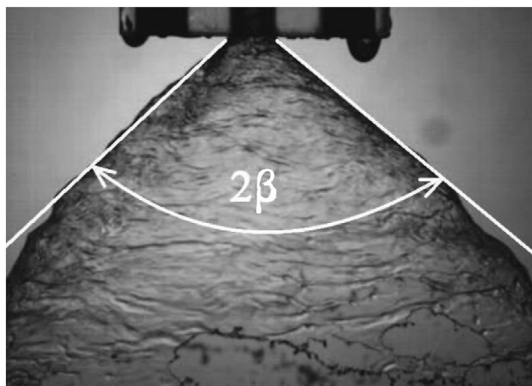
Table 5

Equations of the spray cone angle.

Author	Equation	Type	Injector type
Rizk and Lefebvre [59]	$\cos \beta = \frac{C_d}{K_v(1-X)}, C_d = K_v \left[\frac{(1-X)^3}{1+X} \right]^{0.5}$	Theoretical	
Rizk(Liu et al. [14])	$\cos^2 \beta = \frac{1-X}{1+X}$ $\cos \beta = \frac{C_d}{K_v(1-X)} = \frac{0.35K^{0.5}(D_i/D_0)^{0.25}}{K_v(1-X)}$ $K_v = 0.00367K^{0.29} \left(\frac{\Delta P_i \rho_i}{\mu_i} \right)^{0.2}$	Theoretical	
Liu [67]	$\cos \beta = 0.302(1 + \tan \theta)^{0.414} \left(\frac{1}{A} \right)^{0.35} \times \left(\frac{L_0}{D_0} \right)^{0.043} \left(\frac{D_i}{D_0} \right)^{0.026} + 0.612$	Experimental	Converge-end
Giffen and Muraszew (Santangelo [108], Couto et al. [63] and Jeng et al. [48])	$\sin \beta = \frac{\pi C_d}{2K(1+\sqrt{X})} = \frac{(\pi/2)(1-X)^{1.5}}{K(1+\sqrt{X})(1+X)^{0.5}}$ $C_d = \left[\frac{(1-X)^3}{1+X} \right]^{0.5}, K^2 = \frac{\pi^2(1-X)^3}{32X^2}$	Theoretical	
Xue et al. [15]	$\sin \beta = \frac{\pi C_d \sin \theta}{2K(1+\sqrt{X})} \left(\frac{R_{sw}}{R_s} \right)$ $K^2 = \frac{\pi^2(1-X)^3}{32X^2} \left(\frac{R_{sw}}{R_s} \right)^2 \sin^2 \theta$ $R_{sw} = R_s - R_t$	Theoretical	
Giffen and Muraszew (Liu et al. [22])	$\tan \beta = \frac{2\varphi}{1+\sqrt{1-\varphi}} \times \frac{R_s R_0}{n R_t^2}$	Theoretical	
Orzechowski	$\tan \beta = \frac{2C_d A}{\sqrt{(1+S_b)^2 - 4C_d^2 A^2}}$	Theoretical	
(Chinn [107] and Moon et al. [102])	$S_b = \frac{a_0}{R_0} = \text{Air core diameter at the injector exit/Injector diameter}$		
Fu et al. [71]	$\tan \beta = 0.033 \times A^{0.338} \times Re_t^{0.249}$	Experimental	Open-end
Inamura et al. [57,58]	$\tan \beta = \frac{k}{\sqrt{1-k^2}}$ $\beta_r = \beta \frac{a}{\sqrt{\frac{k}{\cos \beta}}} \exp \left(-\frac{b}{Re} \right)$ $a = 18.9, b = 670, k = \frac{D_{at}}{D_0}$ D_{at} : Air core diameter at the axial position of the tangential ports β_r : Spray half angle at a distance L to the injector exit h_t : Film thickness at the axial position of the tangential ports	Theoretical	
Rizk and Lefebvre (Ma [70], van Banning et al. [109] and	$2\beta = 6 \left(\frac{A_t}{D_i D_0} \right)^{-0.15} \left(\frac{\Delta P_i D_0^2 \rho_i}{\mu_i^2} \right)^{0.11}$	Experimental	Converge-end
Khil et al. [43,44])	$2\beta = 6K^{-0.15} \left(\frac{\Delta P_i D_0^2 \rho_i}{\mu_i^2} \right)^{0.11}$		
Benjamin et al. [69,70]	$2\beta = 9.75 \left(\frac{A_t}{D_i D_0} \right)^{-0.237} \left(\frac{\Delta P_i D_0^2 \rho_i}{\mu_i^2} \right)^{0.067}$	Experimental	Converge-end

penetration. Daviault et al. [110] used two orthogonal lines to the nozzle axis at two axial distances from the injector tip to define four points on the edge of the spray. And then used these four points to define two lines yielding two half angles relative to the injector axis. The sum of the two half angles is defined as the spray cone angle.

For a pressure swirl injector with given geometrical parameters, the spray cone angle is determined by the operation condition. The most important operation parameter is the pressure drop. The spray cone angle increases with the increase of the pressure drop [73,101], and the spray pattern also varies with the increase of pressure drop. It means that

**Fig. 14.** Definition of the spray cone angle [71].

variation of the spray cone angle is related with that of the spray pattern. Ghorbanian et al. [19] found that the spray cone angle increases rapidly with the increase of pressure drop in the onion stage, and it increases slower in the tulip stage and much slower in wavy stage. When the spray is fully developed, the spray cone angle becomes constant and pressure independent. Besides with the pressure drop, liquid viscosity, back pressure and surface tension also have great influence on the spray cone angle. The spray cone angle decreases with the increase of liquid viscosity and back pressure when it becomes pressure independent [18,23,102,113,114]. And if the spray cone angle is small enough, the surface tension converges the spray along the axis [58].

The most important geometrical parameter that influence the spray cone angle is the injector constant K . Any geometrical variation that increase the injector constant K decreases the spray cone angle [48]. For example, the decrease of injector diameter and the increase of the tangential ports area decrease the spray cone angle [32,115]. The increase of the tangential ports number, the geometry characteristics constant A and the tangential ports angle θ all increase the spray cone angle [15,65,73,74]. Other geometrical parameters includes the orifice length and the converge angle of the swirl chamber. The spray cone angle is found to decrease with the increase of the orifice length [75] and the converge angle of the swirl chamber [15,23,74,75,115]. Only Datta and Som [32] found that the spray cone angle increases with the increase of the convergent angle of the swirl chamber. The reason could be that the convergent angle of the swirl chamber in their research is too small that the increase of the convergent angle could significantly decrease the length of the converge section and then decrease the friction loss. Besides

with these geometrical parameters, the trumpet at the injector exit could also influence the spray cone angle, but the influence is still not clear now. Xue et al. [15,74] found that the spray cone angle increases with the increase of the trumpet angle. However, Liu et al. [14] found it decreases with the increase of the trumpet angle because this geometry variation increases the axial velocity and decreases the circumferential velocity of the liquid film.

2.2.4. Breakup length

There are two types of breakup length: slant breakup length L_{bu} and vertical breakup length L_v . The slant breakup length characterizes the distance between the breakup position of liquid film and the injector exit, while the vertical breakup length characterizes the vertical distance between them. These two breakup lengths are related with $L_v = L_{bu} \cos \beta$.

The liquid film is pinched off when the amplitude of the surface wave exceeds a critical value η_{bu} . And the critical amplitude η_{bu} can be calculated by $\eta_{bu} = \eta_0 e^{\omega_s \tau_{bu}}$, where η_0 is the initial amplitude, ω_s is the maximum growth rate of the surface wave which can be calculated by the linear instability analysis, τ_{bu} is the breakup time of the liquid film. Based on this equation, the breakup time of the liquid film can be calculated by:

$$\tau_{bu} = \frac{1}{\omega_s} \ln \left(\frac{\eta_{bu}}{\eta_0} \right) \quad (10)$$

Once the velocity of the liquid film is known, the breakup length can be calculated by:

$$L_{bu} = \tau_{bu} V_0 = \frac{V_0}{\omega_s} \ln \left(\frac{\eta_{bu}}{\eta_0} \right) \quad (11)$$

It is clear that three parameters are needed to calculate the breakup length of liquid film: $\ln(\eta_{bu}/\eta_0)$, the maximum surface wave growth rate ω_s and the liquid film velocity V_0 . First, the liquid film velocity V_0 can be obtained from experiment data or derived from inviscid theory. Second, the maximum surface wave growth rate ω_s can be calculated by the linear instability analysis. However, neither the dispersion equation of conical liquid film nor that of annular liquid film has analytical solution. Thus the analytical solution of a planar film has long been used to predict the breakup length of a conical liquid film. As the breakup of a conical liquid film is dominated by the sinuous waves [81], it is more reasonable to use the analytical solution of this mode of surface waves to calculate the breakup length. Senecal et al. [116] derived the dispersion equations of a viscous planar liquid film with varicose and sinuous waves separately. And obtained the growth rate of the sinuous long waves:

$$\omega_i = -2\nu_i k^2 + \sqrt{4\nu_i^2 k^4 + \frac{2QU^2 k}{h} - \frac{2\sigma k^2}{\rho_l h}} \quad (12)$$

the growth rate of the sinuous short waves:

$$\omega_i = -2\nu_i k^2 + \sqrt{4\nu_i^2 k^4 + QU^2 k - \frac{\sigma k^2}{\rho_l}} \quad (13)$$

The above equations indicate that the growth rate of the sinuous short waves has nothing to do with the film thickness while that of the sinuous long waves is related with the film thickness. Once the analytical solution of the sinuous wave growth rate is derived, the maximum surface wave growth rate ω_s can be obtained. Third, $\ln(\eta_{bu}/\eta_0)$ is an empirical coefficient. Dombrowski and Hooper [117] found that $\ln(\eta_{bu}/\eta_0) = 12$ for planar liquid film agrees well with experiment data, and it has the same value for liquid jets in literature. Since then, this value has been widely used for breakup length prediction. Senecal et al. [116], Moon et al. [102], Laryea and No [118], Inamura et al. [57,58], Hosseinalipour et al. [85], Sivakumar et al. [119], Xiao and Huang [120], Tratnig and Brenn [121] all set $\ln(\eta_{bu}/\eta_0) = 12$. However, Kim et al. [122] found that this value is not a universal constant and should be obtained from experiment. He found that the breakup length predicted with $\ln(\eta_{bu}/\eta_0) = 6.9$ agrees well with experiments. Fu et al. [81] also think that $\ln(\eta_{bu}/\eta_0)$ is not a universal constant, and found that $\ln(\eta_{bu}/\eta_0) = 2.5$ agrees well with experiments. Furthermore, Clark and Dombrowski [123] found that this value is also related with the operation condition. When the liquid Reynolds number is larger than 9000, $\ln(\eta_{bu}/\eta_0) = 12$, otherwise $\ln(\eta_{bu}/\eta_0) = 50$.

The above breakup length prediction is based on the assumption that the liquid film thickness keeps constant along the stream, but in fact the film thickness of a conical liquid film is thinning along the stream, as shown in Fig. 15. Thus the growth rate of long waves is time-dependent, and the time-averaged growth rate is more appropriate for breakup time calculation. Namely

$$\ln \left(\frac{\eta}{\eta_0} \right) = \int_0^t \omega dt \quad (14)$$

As the viscosity has little influence on the growth rate of the long wave [116], Equation (12) can be simplified:

$$\omega_i = \sqrt{2 \frac{QU^2 k^2 - \sigma k^3 / \rho_l}{kh}} \quad (15)$$

where h is the conical liquid film thickness downstream, as shown in Fig. 16. It can be calculated by

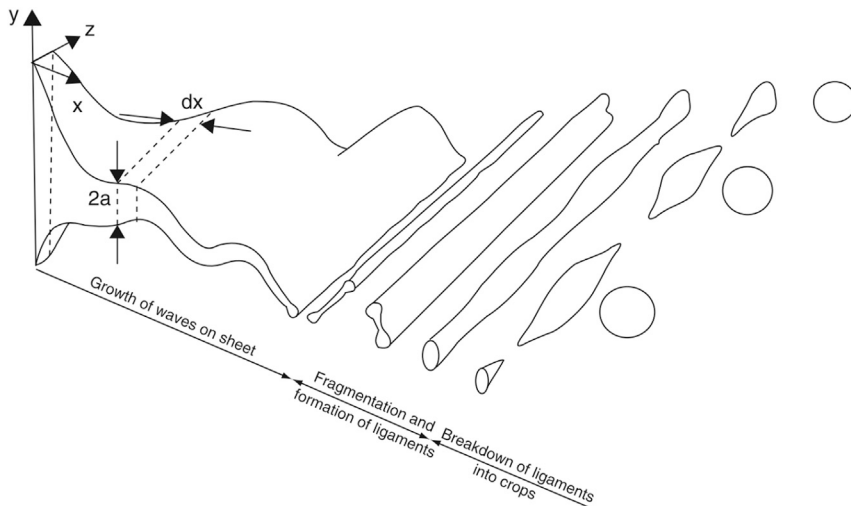


Fig. 15. Schematic of the breakup of a thinning film [124].

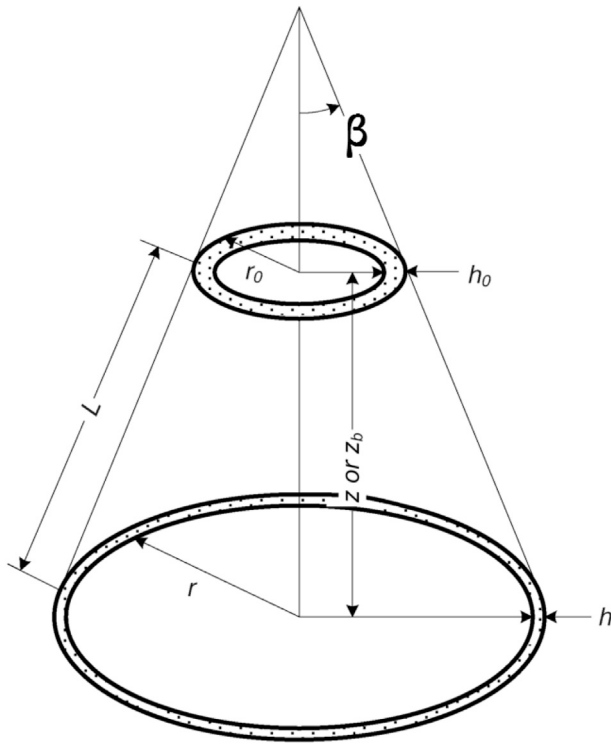


Fig. 16. Schematic of the film thickness of a thinning film [102].

$$h = \frac{h_0(D_0 - h_0)}{D_0 - h_0 + 2V_0 t \sin \beta} \quad (16)$$

Substitute Equation (16) into Equation (15) and integrate Equation (14), then substitute the results into Equation (11) and obtained the breakup length of a conical liquid film:

$$L_{bu} = \frac{1}{2 \sin \beta} \left(\left(\frac{3}{2} \ln \left(\frac{\eta_{bu}}{\eta_0} \right) \sqrt{2} \sin \beta V_0 \sqrt{\frac{\rho_l h_0 (D_0 - h_0)}{QU^2 \rho_l - k\sigma}} + (D_0 - h_0)^{3/2} \right)^{2/3} - D_0 + h_0 \right) \quad (17)$$

where the most unstable wave number $k = \frac{QU^2 \rho_l}{2\sigma}$.

It is clear that the conical liquid film is thinning film. The film thinning influences the growth rate of surface waves and makes the prediction equation different from that of planar liquid film. The breakup length prediction equation (17) also contains some parameters that should be obtained from experiment. In literature, theoretical derivations and empirical coefficients are bond to predict the breakup length, and these semi-empirical equations are listed in Table 6.

These semi-empirical equations indicate that the breakup length is strongly related with the flow characteristics. And the flow characteristics include the liquid film velocity, the gas-liquid relative velocity, the backpressure, the gas density, the liquid density, the surface tension and the spray cone angle. A larger liquid film velocity or gas-liquid relative velocity, a larger backpressure or gas density and a larger spray cone angle would produce smaller breakup length. And a larger liquid density or surface tension would produce a thicker liquid film at the injector exit, which in turn increase the breakup length. In experiment, these flow

Table 6
Equations of the breakup length.

Author	Equation	Film type	Wave mode
Senecal et al. [116]	$L_{bu} = V_0 \left[3 \ln \left(\frac{\eta_{bu}}{\eta_0} \right) \right]^{2/3} \left(\frac{J\sigma}{Q^2 U^4 \rho_l} \right)^{1/3}$ $h = J/t$, J is constant, $\ln(\eta_{bu}/\eta_0) = 12$	Thinning film	Long wave
Kim et al. [122]	$\tau_{bu} = \frac{R_0}{V_0 \tan \beta} \left[\left(\frac{C h_0 \tan \beta}{R_0} \left(\frac{\rho_g}{\rho_l} \right)^{-1} We_l^{-1/2} \right)^{2/3} - 1 \right]$ $\left(\frac{R_0}{h_0 \tan \beta} \right) \left[\left(\frac{L_{bu} \tan \beta}{R_0} + 1 \right)^{3/2} - 1 \right] = C \left(\frac{\rho_g}{\rho_l} \right)^{-1} We_l^{-1/2}$ C is a constant that is related with $\ln(\eta_{bu}/\eta_0)$ $\ln(\eta_{bu}/\eta_0) = 6.9$ correspond to $C = 14.78$	Thinning film	
Clark and Dombrowski [123]	$L_{bu} = C \times \left[\frac{\rho_l \sigma K \ln(\eta_{bu}/\eta_0)}{\rho_g^2 U^2} \right]^{1/3}$, $K = hx$	Thinning film	Long wave
Moon et al. [102]	$L_{bu} = \frac{1}{2 \sin \beta} \left[3 V_0 \sin \beta \ln \left(\frac{\eta_{bu}}{\eta_0} \right) \sqrt{\frac{h_0 (D_0 - h_0)}{QU^2 k - \sigma k^2 / \rho_l}} \right]^{2/3} + (D_0 - h_0)^{3/2}$ $\ln(\eta_{bu}/\eta_0) = 12$	Thinning film	Long wave
Sivakumar et al. [119]	$L_{bu} = \frac{1}{2 \sin \beta} \left\{ \left[\frac{3 \sqrt{2} V_0 \sin \beta \ln \left(\frac{\eta_{bu}}{\eta_0} \right)}{\times \sqrt{\frac{\sigma h_0 (D_0 - h_0) \cos \beta}{Q^2 U^4 \rho_l}}} \right]^{2/3} + (D_0 - 2h_0)^{3/2} - (D_0 - 2h_0) \right\}$ $\ln(\eta_{bu}/\eta_0) = 12$	Thinning film	Long wave
Fu et al. [81]	$L_{bu} = 0.82 \left[\frac{\rho_l \sigma \ln(\eta_{bu}/\eta_0) h_0 \cos \beta}{\rho_g^2 U^2} \right]^{0.5}$ $\ln(\eta_{bu}/\eta_0) = 2.5$	Conical film	
Inamura et al. [57,58]	$L_{bu} = 0.2175 \left[\frac{\rho_l \sigma \ln(\eta_{bu}/\eta_0) h_0 \cos \beta}{\rho_g^2 U^2} \right]^{0.3}$	Conical film	

$\ln(\eta_{bu}/\eta_0) = 12$, U : gas liquid relative velocity.

characteristics are influenced by the operation condition and the geometrical parameters of the injector. The operation conditions mainly influence the liquid film velocity V_0 or the gas-liquid relative velocity U , the gas density, the spray cone half angle β and the film thickness at the injector exit h_0 . A lot of experiments show that the increase of the liquid viscosity increases the breakup length [18,113,114]. And the increase of the backpressure decreases the breakup length [122,125,126], because the intensified aerodynamic force increases the amplitude of the surface waves, which in turn promote the breakup of the liquid film [122]. The injector geometrical parameters also influence the liquid film velocity V_0 or the gas-liquid relative velocity U , the spray cone half angle β and the film thickness at the injector exit h_0 . The larger the geometry characteristics constant, the larger the breakup length [65].

Besides with the flow characteristics, the spray patterns and the turbulence in the liquid film also have great influence on the breakup length. Ghorbanian et al. [19] found that the breakup length increases with the increase of pressure drop in the onion stage, and then decreases with the increase of pressure drop in the next stages. Inamura et al. [58] found that the breakup length increases with the increase of liquid

velocity when the flow is laminar and decreases with that when the flow is turbulent.

2.3. Atomization characteristics of the conical liquid film

Atomization of the conical liquid film is quite complicate, which include primary atomization and secondary atomization. The primary atomization defines the breakup of the liquid film and the further breakup of the produced ligaments. And the secondary atomization defines the breakup of the big droplets. Typical spray structure of pressure swirl injector is shown in Fig. 17.

Atomization characteristics are the parameters that characterize the atomization efficiency. And they are the most direct indexes to evaluate the performance of the pressure swirl injector. The atomization characteristics include the diameter distribution, the velocity distribution, the diameter-velocity distribution and the mass flow rate distribution. Limited by the experimental technique, it is hard to obtain the diameter and velocity of all the droplets in the spray at a specific time. Thus the diameter, velocity and mass flow rate are statistical results of the droplets

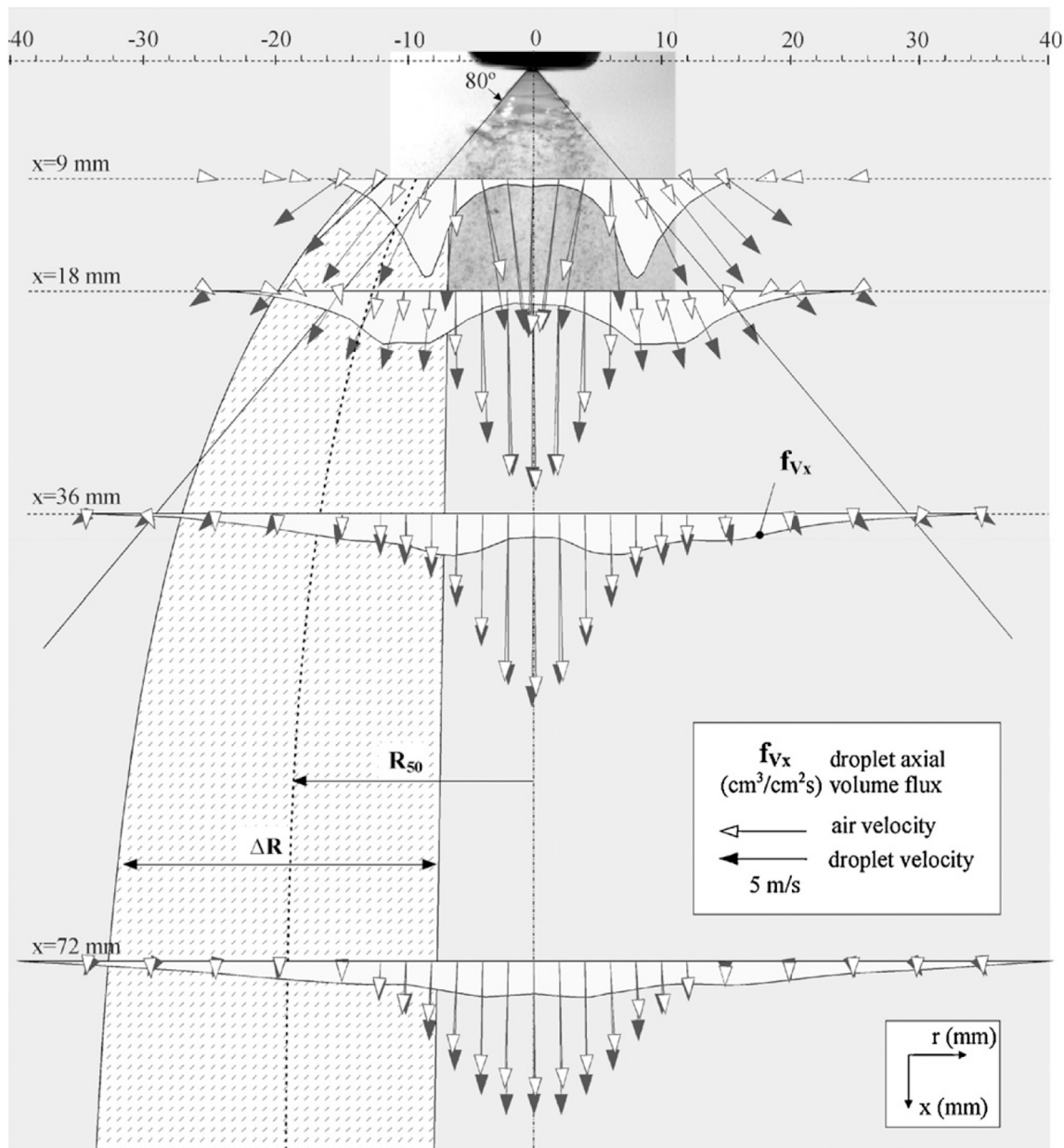


Fig. 17. Typical spray structure of pressure swirl injector [127].

accumulated over some time at one point in the spray. And these parameters are the space distributions of the statistical diameter, the statistical velocity and the statistical mass flow rate.

2.3.1. SMD distribution

The conical liquid film first breakup into ligaments, and then these ligaments further breakup into droplets. Assume that the produced droplets have the same diameter, thus the theoretical diameter of the droplets d can be calculated with the diameter of the ligaments d_l . However, the practical atomization process is much more complicate than the theoretical case, because the droplets with different diameters can be produced by multiple ways. For example, the intense gas liquid interaction can peel droplets from the film, the turbulence energy can shake out droplets from the film and the satellite droplets are produced accompany with the production of ligaments. So, the theoretical analysis is insufficient to reflect the practical diameter of the spray. And experiment is still the most important approach to obtain the statistical diameter of the spray.

Sauter mean diameter (SMD) and its' distribution is the most important parameter that characterize the statistical diameter of the spray. And the definition of SMD is:

$$\text{SMD} = \frac{\sum D_i^3 N_i}{\sum D_i^2 N_i} \quad (18)$$

The SMD can be measured with Mie scattering technique (Malvern sizer), Phase Doppler Anemometry (PDA), laser holography and image processing of instantaneous spray. And empirical equations of SMD is then obtained with these measurement methods. However, the

application of these empirical equations is limited because these equations are strongly related with the geometry and machining accuracy of the given pressure swirl injector. To overcome the deficiencies of theoretical analysis and experimental derived prediction equations, both theoretical analysis and experimental results are combined together. And a semi-empirical prediction equation of SMD is then derived. All the theoretical, empirical and semi-empirical prediction equations of SMD are listed in Table 7.

It is clear that SMD is significantly influenced by the injector geometry, operation condition and physical parameters of liquid. The injector geometry includes the geometry characteristics constant K , the injector diameter D_0 , tangential ports area A_t , diameter of the swirl chamber D_s and the orifice length L_0 . The operation condition includes the pressure drop ΔP_l and the mass flow rate $\dot{m}_l$. And the physical parameters contain the surface tension σ , liquid density ρ_l , ambient gas density ρ_g and liquid viscosity μ_l . Besides with these geometry parameters, operation parameters and physical parameters, some parameters determined by both injector geometry and operation condition also have great influence on SMD. For example the film thickness at the injector exit h_0 , the spray cone half angle β and the velocity of the liquid film V_0 .

First, the influence of the operation condition on SMD. As the pressure drop ΔP_l and the liquid mass flow rate $\dot{m}_l$ are related with each other, only the variation trend of SMD with pressure drop is given. And the variation trends of SMD with the liquid mass flow rate and liquid velocity are similar with the variation trend of SMD with pressure drop, because the liquid mass flow rate $\dot{m}_l$ and liquid film velocity V_0 increase with the increase of pressure drop ΔP_l . The results through extensive experiment show that the SMD decreases with the increase of pressure drop [19,63,

Table 7
SMD calculation equations.

Author	Equation	Method
Sivakumar et al. [119]	$d_l = \sqrt{\frac{4h_0}{K_s}}$, where $K_s = \frac{\rho_g V_0^2}{2\sigma}$	Theoretical
Moon et al. [102]	$d = 1.88d_l(1 + 3Oh)^{\frac{1}{5}}$, where $Oh = \frac{\mu_l}{\sqrt{\rho_l \sigma d_l}}$	
Dombrowski and	$d_l = 0.9614 \left(\frac{k_0^2 \sigma^2}{\rho_g \rho_l V_0^2} \right)^{1/6} \left[1 + 2.6\mu_l \left(\frac{k_0 \rho_g^4 V_0^2}{72\rho_l^2 \sigma^5} \right)^{1/3} \right]^{1/5}$	Theoretical
Johns(Xiao and Huang [120])	$d = 1.88d_l(1 + 3Oh)^{\frac{1}{5}}$, $Oh = \frac{\mu_l}{\sqrt{\rho_l \sigma d_l}}$, $k_0 = h_0 x$	
Couto et al. [63,128]	$d_l = 0.9614 \cos\beta \left(\frac{h_0^4 \sigma^2}{\rho_g \rho_l V_0^4} \right)^{1/6} \times \left[\left(\frac{h_0^2 \rho_g^4 V_0^2}{72\rho_l^2 \sigma^5} \right)^{1/3} \right]^{1/5}$	Theoretical
Radcliffe(Xiao and Huang [120], Santangelo et al. [129])	$d = 1.89d_l$	Empirical
Jasuja(Xiao and Huang [120], Yang [130])	$\text{SMD} = 7.3\sigma^{0.6}\mu_l^{0.2}\rho_l^{0.25}\rho_g^{0.25}\Delta P_l^{-0.4}$	Empirical
Ballester(Xiao and Huang [120])	$\text{SMD} = 4.4\sigma^{0.6}\mu_l^{0.16}\rho_l^{0.16}\rho_g^{0.22}\Delta P_l^{-0.43}$	Empirical
Lefebvre(Santolaya et al. [103], Park and Heister [131])	$\text{SMD} = 0.436\mu_l^{0.55}\Delta P_l^{-0.74}D_0^{-0.05}A_t^{-0.24}$	Empirical
Van Banning et al. [109]	A_t : Area of the tangential ports $\text{SMD} = 2.25\sigma^{0.25}\mu_l^{0.25}\rho_l^{0.25}\rho_g^{-0.25}\Delta P_l^{-0.5}$	Empirical
Liu et al. [14]	$\text{SMD} = \left(\frac{\sigma\mu_l\dot{m}_l}{\rho_g} \right)^{0.25} (3.29 \pm 0.06) \times 10000 \times \Delta P_l^{-0.5}$	Empirical
Wang and Lefebvre (Sivakumar et al. [119], Semiao [132])	$\text{SMD} = 0.5536\sigma^{0.25}\mu_l^{0.25}\rho_l^{0.125}D_0^{0.5} \times \rho_g^{-0.25}\Delta P_l^{-0.375}f\left(\frac{D_s}{D_0}, \frac{L_0}{D_0}, \beta\right)$ where $f = \left(\frac{D_s}{D_0}\right)^{0.33} \left(\frac{L_0}{D_0}\right)^{0.122} (1 + \tan\beta)^{1.38}$	Empirical
Wang and Lefebvre (Xiao and Huang [120], Santangelo [108])	$\text{SMD} = A \left(\frac{\sigma\mu_l^2}{\rho_g \Delta P_l} \right)^{0.25} (h_0 \cos\beta)^{0.25} + B \left(\frac{\sigma\mu_l}{\rho_g \Delta P_l} \right)^{0.25} (h_0 \cos\beta)^{0.75}$ where $A = 2.11[\cos 2(\beta - 30)]^{2.25} \left(\frac{3.4 \times 10^{-4}}{D_0} \right)^{0.4}$ $B = 0.635[\cos 2(\beta - 30)]^{2.25} \left(\frac{3.4 \times 10^{-4}}{D_0} \right)^{0.2}$	Empirical
Davanlou et al. [133]	$\text{SMD} = 4.52 \left(\frac{\sigma\mu_l^2}{\rho_g \Delta P_l} \right)^{0.25} (h_0 \cos\beta)^{0.25} + 0.39 \left(\frac{\sigma\mu_l}{\rho_g \Delta P_l} \right)^{0.25} (h_0 \cos\beta)^{0.75}$	Empirical
Xiao and Huang [120]	$\text{SMD} = 2.11 \left(\frac{\sigma\mu_l^2}{\rho_g \Delta P_l} \right)^{0.25} (h_0 \cos\beta)^{0.25} + 0.62 \left(\frac{\sigma\mu_l}{\rho_g \Delta P_l} \right)^{0.1} (h_0 \cos\beta)^{0.9}$	Empirical
	$\text{SMD} = 22 \left[\frac{KD_0(1+\sqrt{X})}{1-X} \right]^{\frac{1}{2}} \left[\frac{h_0^{1.17}(D_0-h_0)^{0.67}}{m_l^{0.67}} \right] \left(\frac{\sigma^2 \rho_l^2}{\rho_g} \right)^{\frac{1}{8}}$	Semi-empirical

109,120,129,130,134–140]. It is because that the increase of pressure drop can accelerate the liquid film. And the accelerated liquid film have larger turbulence energy and larger growth rate of surface waves, which in turn promote the breakup of conical liquid film. What's more, the increase of pressure drop can decrease the film thickness at the injector exit h_0 and increase the spray cone half angle β , both the variations of h_0 and β promote the breakup of liquid film and decrease SMD [120]. Eberhart et al. [134] found that a 22% mass flow rate sub-nominal throttle increases the surface wave length by 30%, and a 23% mean axial velocity decrease increases the SMD by 9%.

Second, the influence of the physical parameters on SMD. The SMD increases with the increase of surface tension σ and liquid viscosity μ_l , which indicates that the surface tension and liquid viscosity suppress the breakup and atomization of the conical liquid film [114,135]. The reason is that the increase of surface tension suppresses the growth of the surface waves, and the increase of liquid viscosity increases the film thickness at the injector exit and decreases the spray cone angle. Both the increase of film thickness and the decrease of spray cone angle increase the SMD [120]. Besides with the surface tension and liquid viscosity, the temperature of the liquid and the ambient gas density also have great influence on SMD. Kim et al. [136,137,141] and van Banning et al. [109] found that the SMD decreases with the increase of liquid temperature especially when the liquid is superheated, because the increase of liquid temperature decreases the surface tension, which in turn promote the breakup of the conical liquid film. What's more, with the increase of liquid temperature, the perforation breakup seems to be more frequent and distinct, and onset of holes on the conical liquid film occurs earlier [136]. And the perforation also promotes the breakup and atomization of the conical liquid film. The increase of the ambient gas density ρ_g decreases SMD, because it intensify the gas liquid interaction and promote the breakup of the conical liquid film.

Third, the influence of the injector geometry on SMD. The most important geometrical parameter is the injector constant K , and it is related with the tangential ports area A_t . Xiao and Huang [120] found that SMD increases with the increase of tangential ports area, because the increase of tangential ports area increases the injector constant. And the film thickness at the injector exit increases and the spray cone angle decreases with the increase of injector constant. What's more, the smaller the spray cone angle and the thicker the film at the injector exit, the larger the SMD [120]. On the effects of the orifice length L_0 , Xiao and Huang [120] and Liu et al. [14] found that SMD increases with the increase of orifice length, because it decreases the spray cone angle and increases the film thickness at the injector exit. No matter how the geometrical parameters vary, as long as it decreases the film thickness at the injector exit and increases the spray cone angle, it decreases the SMD.

SMD is a kind of statistical diameter of the spray. And each spray can be characterized by one SMD, but it not means that the droplets in the spray is uniform. In fact, the droplets in the spray are quite nonuniform and have a diameter distribution. Typical diameter distribution of a spray is shown in Fig. 18. There are three methods to model the droplet

diameter distribution: the Maximum Entropy (ME) method, the Discrete Probability Function (DPF) method and the empirical method. Babinsky and Sojka [142] have thorough reviewed the modeling methods of drop size distribution. The maximum entropy method have been widely investigated by Dumouchel et al. [28,143–148], and Mondal et al. [149,150]. The idea of the discrete probability function method is simple. First, it assumes that the conical liquid film initially breakup into ligaments, which eventually collapse into drops. This process is deterministic, and the diameter of the ligaments and droplets can be calculated with the linear analysis. Then, the diameter distribution is produced when the input initial conditions (pressure drop, mass flow rate, velocity, physical properties, etc.) fluctuate with a probability density function. The empirical method is the classical method to model the diameter distribution. It uses a curve to fit to the data obtained in a wide range of operation condition with all kinds of injectors. Babinsky and Sojka [142] summarized seven empirical distributions, as listed in Table 8.

Among these three droplets diameter distribution prediction methods, the empirical method is hard to apply to other injectors, because the distribution function is strongly related with the injector geometry and operation condition. The maximum entropy method focus on the initial and final states of the spray and neglect the primary breakup and secondary atomization. The discrete probability function method is suitable for the spray produced by the primary breakup. The spray with strong gas-liquid interaction is dominated by secondary atomization, and is more appropriate to use the maximum entropy method. As the present paper places emphasis on the physical process of primary breakup and atomization, the maximum entropy method is not discussed too much. For interested readers, Dumouchel's research is recommended.

If the measuring volume is small enough, the SMD calculated with the droplets in this volume can represent the diameter of droplets at this 'point' of the spray. Thus the SMD distribution along radius and axis can be obtained by moving the measuring volume. A large amount of experimental results indicate that the SMD distribution of the pressure swirl injector shows a 'single peak' shape [21,102,103,127,134,139,151–153], as shown in Fig. 19. It is because that the droplets around the stream line of the conical liquid film are directly breakup from ligaments, while the droplets in the center or at the periphery of the spray are entrained from the stream-line region of the liquid film. Secondary atomization occurs during the entrainment, which produces smaller droplets. Normally, the SMD decreases along the axis. But if the spray is tense enough, the SMD initially decreases and then increases slightly along the axis [108,137], because the coalescence of droplets due to collision is significant [133], as shown in Fig. 20. Santangelo [108] found that the secondary atomization governs the droplet size until a critical location is reached, then the coalescence of the droplets increases the diameter of the droplets. Sivakumar et al. [119] found that the liquid film undergoes primary breakup and secondary atomization until 35–45 mm downstream the injector exit. After that the SMD increases for the coalescence of the droplets. Santolaya et al. [103,127,153] investigated the effects of droplets collision on the diameter distribution. And found that the maximum collision rate occurs at the densest zone where the relative velocity of droplets is also high. He figured out that the increase of pressure drop increases the collision rate and the increase of collision Webber number promote the separation of droplets with satellite droplet formation.

2.3.2. Velocity distribution

Droplet velocity is another important parameter of the atomization characteristics. Similar with the SMD, the mean axial velocity of droplets along radius shows a 'single peak' shape [19,21,127,151,153], as shown in Fig. 21. It is clear that the droplet's velocity is negative at the spray center near the injector exit, which indicates that there exists a recirculation zone [134]. The minimum mean axial velocity of the droplets occurs at the center of the spray because these droplets are entrained by the recirculation zone. And the maximum mean axial velocity of the droplets occurs around the stream line of the liquid film because these

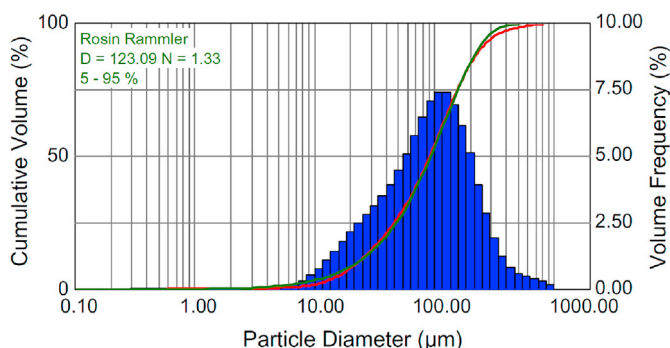


Fig. 18. Typical diameter distribution of a spray [109].

Table 8
Empirical diameter distributions.

Distribution name	Distribution function	Distribution type
The log-normal distribution	$f_0(D) = \frac{1}{D(\ln\sigma_{LN})\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left[\frac{\ln(D/\bar{D})}{\ln\sigma_{LN}}\right]^2\right\}$ $\bar{D}$: logarithmic mean size of the distribution, σ_{LN} : width of the distribution	Number distribution
The upper-limit distribution	$f_3(D) = \frac{\delta D_{max}}{D(D_{max}-D)\sqrt{\pi}} \exp\left\{-\delta^2\left[\ln\left(\frac{aD}{D_{max}-D}\right)\right]^2\right\}$ $a = \frac{D_{max}}{\bar{D}}$, $\delta = \frac{1}{(\ln\sigma_{UL})\sqrt{2}}$ $\bar{D}$: representative diameter, D_{max} : maximum drop diameter, σ_{UL} : distribution width	Volume distribution
The root-normal distribution	$f_3(D) = \frac{1}{2\sigma_{RN}\sqrt{2\pi D}} \exp\left\{-\frac{1}{2}\left[\frac{\sqrt{D}-\sqrt{\bar{D}}}{\sigma_{RN}}\right]^2\right\}$ $\bar{D}$: mean diameter, σ_{RN} : distribution width	Volume distribution
The Rosin-Rammler distribution	$f_0(D) = q\bar{D}^{-q}D^{q-1}\exp\{- (D/\bar{D})^q\}$ $f_3(D) = 1 - \exp\{- (D/\bar{D})^q\}$ $\bar{D}$: mean diameter, q : distribution width	Number distribution Volume distribution
The Nukiyama-Tanasawa distribution	$f_0(D) = aD^p\exp\{-bD^q\}$, b, p, q adjustable parameters, a is a normalizing constant	Number distribution
The Log-hyperbolic distribution	$f_0(x; \alpha, \beta, \delta, \mu) = a(\alpha, \beta, \delta, \mu) \times \exp\{-\alpha\sqrt{\delta^2 + (x-\mu)^2} + \beta(x-\mu)\}$ $a = \frac{\sqrt{\alpha^2 - \beta^2}}{2\alpha K_1(\delta\sqrt{\alpha^2 - \beta^2})}$ is a normalizing constant, K_1 is the modified Bessel function of the third kind and first order, δ is the scale parameter, μ is the location parameter, α and β describes the shape of the PDF	Number distribution
The three-parameter log-hyperbolic distribution	$f_0(x) = A \exp\left\{\frac{a}{a^2\cos^2\theta - \sin^2\theta} \times \sqrt{(a^2\cos^2\theta - \sin^2\theta)(x + \mu_0 - \mu)^2} - \frac{(a^2+1)\sin\theta\cos\theta}{a^2\cos^2\theta - \sin^2\theta} (x + \mu_0 - \mu)\right\}$ where α, θ, μ are shape parameters, A is the normalizing constant $\mu_0 = -\frac{(a^2+1)\sin\theta\cos\theta}{\sqrt{\frac{a^2-(a^2+1)^2\sin^2\theta\cos^2\theta}{a^2\cos^2\theta - \sin^2\theta}}} A = \frac{\sqrt{a^2-(a^2+1)^2\sin^2\theta\cos^2\theta}}{2a\sqrt{a^2\cos^2\theta - \sin^2\theta}K_1\sqrt{\frac{a^2-(a^2+1)^2\sin^2\theta\cos^2\theta}{a^2\cos^2\theta - \sin^2\theta}}}$	Number distribution

droplets directly breakup from the ligaments and inherit the film velocity.

2.3.3. Diameter-velocity distribution

Besides with SMD and mean axial velocity, diameter-velocity distribution is also an important atomization characteristics. It shows the corresponding relation of the droplet diameter and the droplet velocity. At present research on diameter-velocity distribution is seldom, Belhade et al. [154] and Santolaya et al. [103] shows the diameter-velocity distribution in there research, and the typical diameter-velocity distribution measured by PDA is shown in Fig. 22. Ayres et al. [155] proposed an prediction model for diameter-velocity distribution based on the maximum entropy formalism, and the typical predicted diameter-velocity distribution is shown in Fig. 23. The predictions agree well with the available data for the velocity distribution.

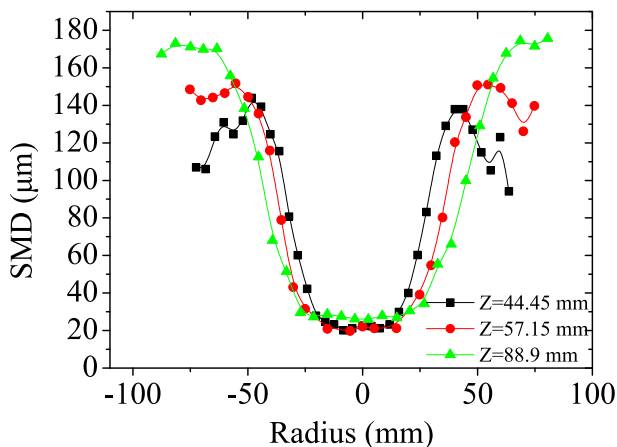


Fig. 19. Typical SMD distribution along radius [134].

2.3.4. Mass flow rate distribution

The mass flow rate distribution have great influence on the mixing, which in turn influence the combustion. Similar with the distribution of SMD and mean axial velocity, the mass flow rate distribution (or the mass/volume flux) shows ‘single peak’ shape, as shown in Fig. 17. It is because that the primary breakup and secondary atomization produces a large amount of droplets around the stream line of the liquid film. And these droplets have larger SMD and mean axial velocity than the center and the periphery of the spray. Santolaya et al. [103,127,153] found that the liquid flow rate is concentrated in a small region where the droplets presented a high dispersion of sizes. And the high collision rates of droplets in the dense spray could redistribute the liquid mass flow rate. Koh et al. [156] measured the mass flow rate from laser induced

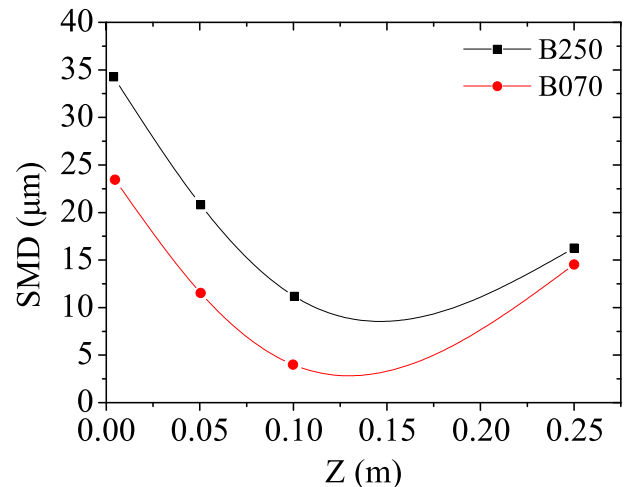


Fig. 20. Typical SMD distribution along axis [108].

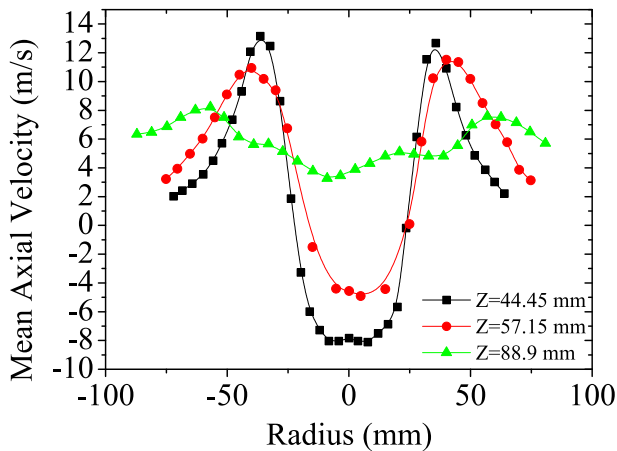


Fig. 21. Typical mean axial velocity distribution of the pressure swirl injector [134].

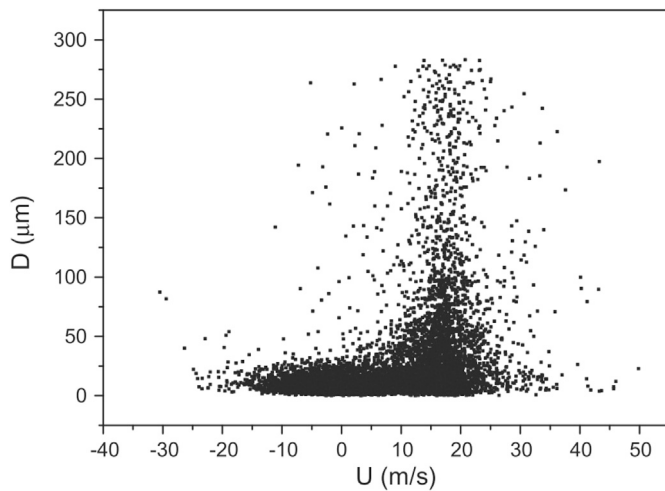


Fig. 22. Typical diameter-velocity distribution of droplets in one measuring volume [154].

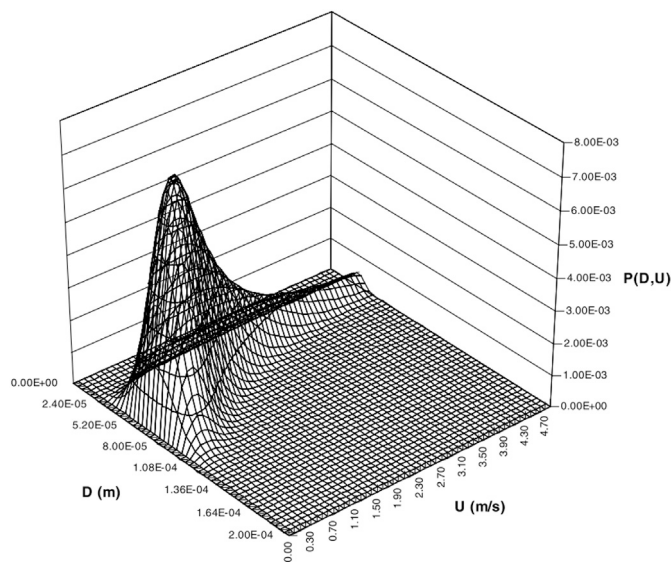


Fig. 23. Diameter-velocity distribution predicted by the theoretical model [155].

fluorescence images, and found it agrees well with the mass flow rate measured by PDPA.

2.4. Effects of the rocket engine environment

In the liquid rocket engine, the chamber pressure and the combustion gas temperature are really high, which indicates that the atomization occurs with the back pressure and high temperature. If the pressure exceeds the critical pressure and the temperature exceeds the critical temperature, the propellant is supercritical. The atomization of propellant at supercritical conditions is quite different from that at normal condition. In this section, the effects of high back pressure and temperature on the inner flow characteristics of the pressure swirl injector and the atomization of the conical liquid film are reviewed.

2.4.1. Effects of back pressure

First, the effects of back pressure on the internal flow characteristics of pressure swirl injector. Kenny et al. [49,50] found that the increase of back pressure increases the film thickness if the mass flow rate keeps constant. And the corresponding discharge coefficient also increases because the increase of film thickness decreases the axial velocity of the liquid film, thus the friction loss decreases sequentially. He also found that the back pressure significantly influence the internal swirling flow that several “necks” arise on the air core, and the dimension and location of these necks vary with the back pressure and tends to a constant. Chen et al. [157] simulated the internal flow of pressure swirl injector with back pressure, and found that the increase of film thickness with the back pressure is caused by the intensified gas liquid shear stress that changes the velocity profile. He also captured the stationary surface waves that looks like the necks in experiment, and the location of the stationary surface waves agree well with that of the necks.

Second, the effects of back pressure on the breakup of the conical liquid film. The back pressure increases the gas density and enhance the gas liquid interaction, which significantly promote the primary breakup of the liquid film, as shown in Fig. 24. The experimental results of Kenny et al. [49,50] show that the spray cone angle decreases with the increase of the back pressure. The simulation results of Chen et al. [157] also show that the back pressure converges the conical liquid film and decreases the spray cone angle. What's more, the research of Moon et al. [102], Fu and Yang [125] and Eberhart et al. [126] all indicates that the spray cone angle decreases with the increase of the back pressure. Kim et al. [122] further figured out that the back pressure has little influence on the spray cone angle before the breakup of the conical liquid film and significantly decreases the spray cone angle after that. Although the back pressure has little influence on the cone angle of the liquid film, it greatly affects the primary breakup for the enhancement of the gas liquid interaction. Moon et al. [102] found that the breakup of the conical liquid film is dominated by long wave mode at atmospheric pressure, and is dominated by short wave mode with back pressure, because the back pressure increases the Webber number and enhances the gas liquid interaction. Kim et al. [122] found that the amplitude of the disturbance increases and the breakup length decreases with the increase of the liquid Webber number and back pressure, because the aerodynamic force increases. All the research of Moon et al. [102], Fu and Yang [125] and Eberhart et al. [126] show that the breakup length decreases with the increase of back pressure, indicating that the back pressure promotes the primary breakup.

Third, the effects of back pressure on the atomization characteristics. Eberhart et al. [126] found that the mean axial velocity and droplet size decrease with the increase of back pressure in the primary breakup region. Elbadawy et al. [158] found that it is the increased turbulence level of the ambient gas decreases the SMD, increases the radial penetration of the vapor and increases the droplets number.

2.4.2. Effects of temperature

In liquid rocket engines, the temperature of the propellants in the manifolds are smaller than the critical point. And the propellants is

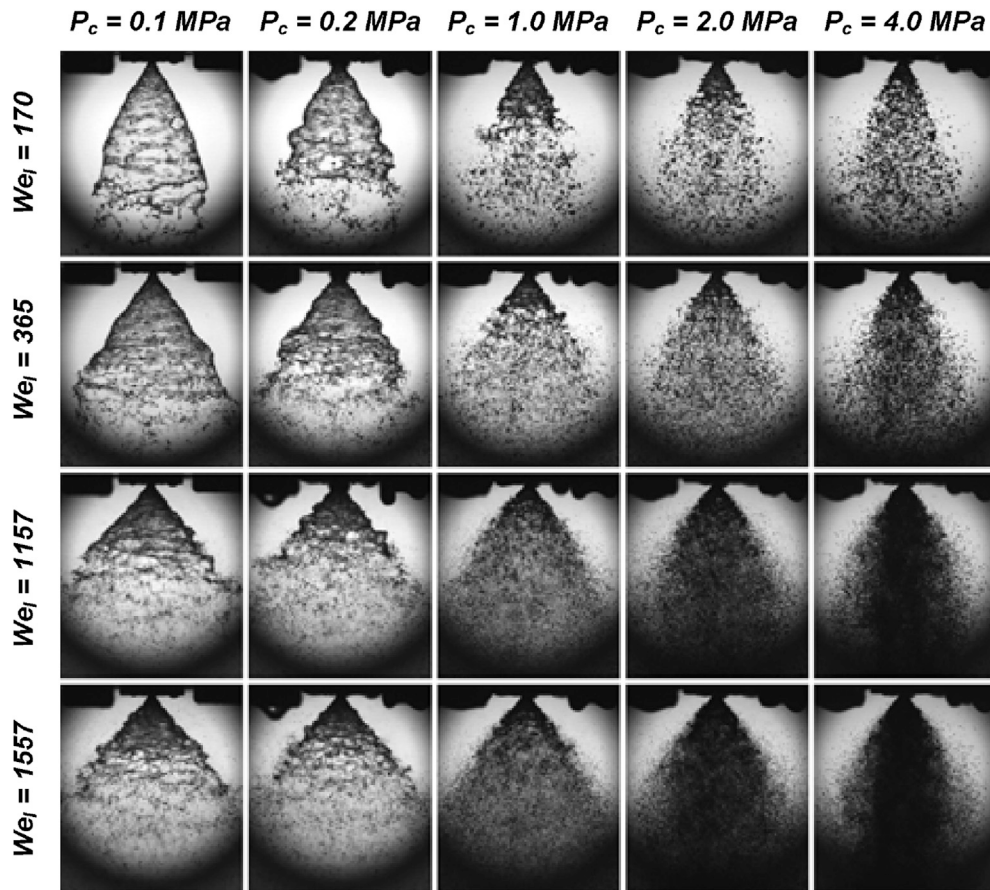


Fig. 24. Effects of back pressure on the breakup of the conical liquid film [122].

heated by the hot gas in the combustion chamber during injection, and may exceed the critical temperature. When the temperature and pressure in the combustion chamber exceeds the critical point of the propellants, the pressure swirl injector works at supercritical condition. It means that the atomization may occur from subcritical condition to supercritical condition. Thus the temperature has significant influence on the atomization of pressure swirl injector.

The supercritical fluid does not exist distinct liquid and gas phases, and does not have a gas-liquid interface. Thus there is no surface tension in a supercritical fluid. By changing the pressure and temperature of the fluid, the properties can be “tuned” to be more liquid-like or more gas-like. And close to the critical point, small changes in pressure or temperature result

in large changes in density. It means that the injection of supercritical propellants is quite different from that of liquid propellants. The conical liquid film disappears and the injection is more like a gas injection. The typical subcritical spray and supercritical spray of a pressure swirl injector are shown in Fig. 25. Cho et al. [159] studied the surface instability of cryogenic swirl flow at subcritical to supercritical conditions. And found that the spray cone angle, the wavelength and the propagation velocity are drastically changed by the phase change. He figured out that at subcritical condition the instability of the cryogenic swirl flow is dominated by the precessing vortex core in the central recirculation zone. And the increase of back pressure decreases the flow velocity, which in turn decreases the frequency of the instability. However, at subcritical to

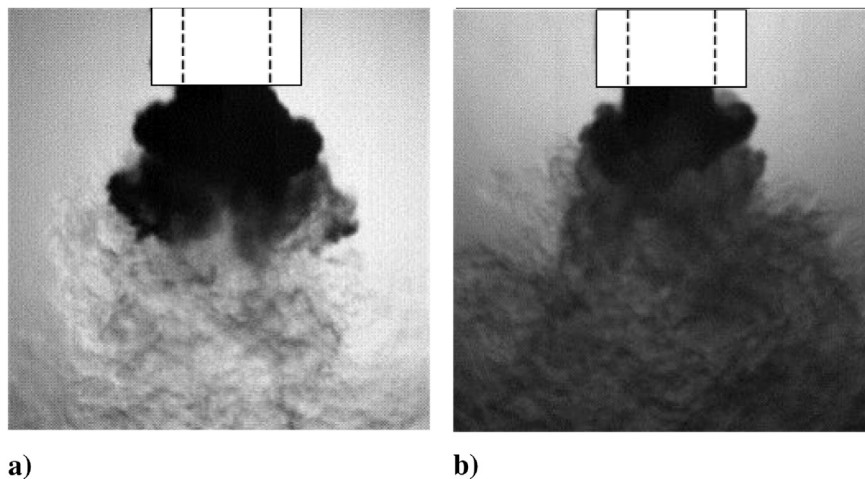


Fig. 25. Instantaneous spray images, a) subcritical condition, b) supercritical condition [159].

supercritical condition, the frequency do not change drastically because the instability of the sub-to-supercritical flow is determined by the flow in a liquid state.

2.5. Response of injector and atomization on the pressure oscillation

When the combustion instability occurs, either the pressure in the supply system or the pressure in the combustion chamber oscillates. These pressure oscillations could induce the spray oscillation and then couple with the combustion. In order to eliminate the combustion instability, the injector dynamics and the response characteristics of the primary breakup and atomization on the pressure oscillation are needed.

2.5.1. Injector dynamics

The pressure oscillation in the supply system could transmit in the injector, causing the film thickness and velocity at the injector exit fluctuate. And the film thickness and velocity oscillation will further influence the primary breakup of the conical liquid film. To understand the effects of the pressure oscillation on the primary breakup, the injector dynamics that describe the response characteristics of the parameters at the injector exit to the pressure oscillation at the injector inlet are needed.

In the theoretical research, Bazarov [160] conducted the pioneer works. He developed a linear analytic model based on the long-wave disturbances traveling on the free surface of the injector core, and analyzed the dynamics of a classical pressure swirl injector. Fu [66] found that the linear model is suitable for the low frequency combustion instability, because the disturbance propagation velocity is obtained by regarding the disturbance waves as long waves. This is true only when the disturbance wavelength is much larger than the injector length. When high frequency combustion instability occurs, the disturbance waves cannot be regarded as long waves, because the disturbance wavelength is in the same level with the injector size. And the disturbance propagation velocity is related with the frequency. Fu [66] corrected the disturbance propagation velocity by including the dispersion equation of the disturbance waves. Based on the corrected model, he investigated the dynamics of an open end swirl injector, and obtained the transfer function of the injector.

In the numerical research, the research group of Heister [131, 161–167] in Purdue university have conducted a lot of simulation on the nonlinear dynamic response of the pressure swirl injector with the boundary element method (BEM). Ismailov et al. [161,162,165,166,168] dissected the entire unsteady flow in swirl injectors into the disturbance wave refraction, wave reflection, and vortex chamber resonance, and conducted the linear small perturbation analysis based on these assumptions. He also conducted the dynamic simulation with the nonlinear boundary element method and found that the dynamic responses peak at the frequency that are coincident with the resonant frequency. Richardson et al. [79,167] compared the simulation results of boundary element method with that of Bazarov's linear model, and found that they agree well when the frequency is smaller than 5000 Hz, after that the deviation occurs.

In the experimental research, two special techniques are required. One is the pulsator that induce the pressure fluctuation, the other is the quantification of the transient mass flow rate at the injector exit. At present hydromechanical pulsator is widely used, Wilson et al. [169,170] and Bazarov et al. [171] have reviewed and investigated the characteristics of the hydromechanical pulsators, and compared the performance. The transient mass flow rate at the injector exit cannot be measured directly, and the most used method is to measure the transient film thickness at the injector exit and then calculate the transient mass flow rate with the transient film thickness and the average film velocity. Fu et al. [39,41] found that the frequency of the film thickness fluctuation coincides with that of the pressure oscillation. And the amplitude of the film thickness fluctuation decreases with the increase of the pressure oscillation frequency. He also figured out that the phase contrast between

the mass flow rate fluctuation and the pressure oscillation increases linearly with the increase of the pressure oscillation frequency. What's more, the phase contrast increases with the increase of the geometry characteristics constant A , and decreases with the increase of the pressure drop. Ahn et al. [172,173] found that the water hammer effects exists when the pressure oscillates, and it affects the resonant peak to deviate the theoretical value. And if the water hammer effects is accounted, the measured resonant peak agrees well with the theoretical value. Jin and Yoon [174] found that the gain of the normalized transfer function increases with the oscillation frequency while the phase contrast of that decreases. However, the gain decreases with the oscillation frequency at around 100–130 Hz because the acoustic characteristics of the fuel feed line may control the flow. Khil et al. [42] measured the pressure, film thickness, axial velocity and mass flow rate at the injector exit, and found that the phase diagrams of the pressure, axial velocity and mass flow rate started from the fourth quadrant and moved in a clockwise direction to quadrants 3, 2, 1. However, the phase diagram of the film thickness started in the first quadrant. He also figured out that the average injection pressure has little influence on the phase diagram, but it decreases the amplitude of the fluctuation because it decreases the relative magnitude of the fluctuated amplitude and the average injection pressure. What's interesting in his research is that the fluctuation amplitude decreases with the increase of oscillation frequency between 0 and 75 Hz, while the opposite trend occurs when the oscillation frequency exceeds 75 Hz. He think that this phenomenon is caused by the Klystron effect.

The injector dynamics can be significantly influenced by the injector geometry. Chung et al. [175] investigated the effects of the diameter and length of the swirl chamber, and found that the mass flow rate oscillation is weakened by the decrease of the swirl chamber diameter and the increase of the swirl chamber length. It indicates that the pressure swirl injector with a slender swirl chamber is dynamically more stable and is better for combustion. The row number of the tangential inlets is another important geometrical parameter. Fu [66] derived the transfer function of a pressure swirl injector with double rows of tangential ports, and found that the oscillation is damped when the distance between the rows equal to half a disturbance wavelength, because the surface waves induced by these two rows of inlet are in opposite phase. Richardson et al. [79,167] also found that the open-end injector with double rows of inlets have better stability characteristics than the injector with a single row of inlets. Zakharov et al. [176] also investigated the mechanism of wave canceling in pressure swirl injector with multi rows of tangential inlets.

As for the open-end swirl injector, Fu et al. [39] found that the amplitude-frequency variation and the phase-frequency variation is similar with that of a classical pressure swirl injector. And the larger the pressure drop, the larger the amplitude and the smaller the phase contrast. He also figured out that the swirl injector that is more slender and has larger geometry characteristic constant A can damp more oscillation energy. Chung et al. [45] investigated the effects of the diameter and length of the swirl chamber on the dynamic characteristics of an open-end swirl injector. He found that the amplitude of the film thickness fluctuation increases with the increase of the pressure oscillation amplitude. And the slope decreases with the increase of the swirl chamber diameter, the swirl chamber length and the tangential inlet number, and decreases with the decrease of the manifold diameter. Furthermore, the amplitude of the liquid film thickness peak at a pulsation frequency of 200 Hz, and the amplitude of this peak decreases with the increase of the swirl chamber diameter and length. Correspondingly, the range of the spray cone angle oscillation peak at the pulsation frequency 200 Hz.

2.5.2. Response of film breakup and atomization on the pressure oscillation

When low frequency combustion instability occurs, the pressure in the supply system oscillates and affects the primary breakup of the conical liquid film, as shown in Fig. 26. Kim and Lai [177] found that the pressure oscillation can promote the growth of the surface waves when

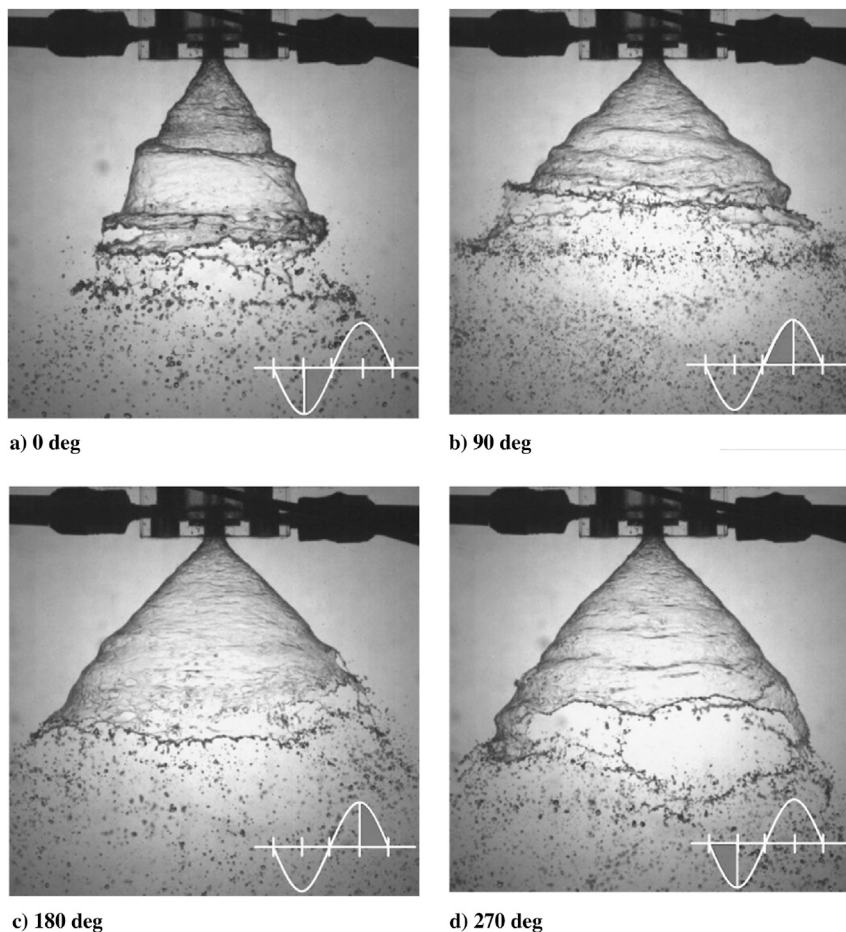


Fig. 26. Different spray patterns with the pressure oscillation in supply system [42].

the pressure drop is low. The spray becomes more uniform, the droplet number density and volume flux increased about 30%, because the primary breakup of the conical liquid film is accelerated by the promoted surface waves.

3. Summary

3.1. Summary: formation of conical liquid film

A lot of researches have focused on the formation of conical liquid film, especially the formation of air core, the growth of boundary layer, the film thickness, the discharge coefficient and the growth of disturbance on the gas liquid interface. These researches are really thorough. However the research on the unstable air core is limited, the mechanism of the double-helical air core is not clear. Thus the unsteady internal flow characteristics should be focused on in the future.

3.2. Summary: breakup of conical liquid film

Research on the breakup of the conical liquid film mainly focus on the spray pattern, the spray cone angle and the breakup length. However, the research, especially the experimental research, on the surface waves which lead to the breakup of the conical liquid film is limited. Only several papers report the investigation on the surface waves with the linear stability analysis.

3.3. Summary: atomization characteristics of the conical liquid film

The atomization characteristics of pressure swirl injector are mainly obtained from experimental research, because the breakup and

atomization of the conical liquid film is very complicate and is very hard to simulate. Researches on the atomization characteristics mainly focus on the SMD and the mean axial velocity distributions, and the researches on the diameter-velocity distribution and the mass flow rate distribution are limited.

3.4. Summary: effects of the rocket engine environment

The rocket engine environment includes the high back pressure and temperature. The back pressure promotes the atomization because it increases the gas density and enhances the gas liquid interaction. High temperature and high pressure may change the propellants from subcritical to supercritical condition. Once the supercritical condition is achieved, the surface tension and the latent heat of evaporation become zero. Thus the combustion is dominated by the mixing state of the propellants. Above all, high back pressure and temperature are good for atomization.

3.5. Summary: response of injector and atomization on the pressure oscillation

The effects of pressure oscillation on the injection and atomization has been investigated for a long time, but researches mainly focus on the injector dynamics. A lot of thorough theoretical, numerical and experimental researches have been conducted, and the response characteristics of the pressure swirl injector have well understood. However, the injector dynamics mainly focus on the response of the mass flow rate at the injector exit on the pressure oscillation in supply system. The effects of the pressure oscillation on the primary breakup and atomization characteristics have not been well researched. The response characteristics of

the primary breakup and secondary atomization on the pressure oscillation is still needed.

4. Conclusion

To ensure the reliability and stability of the liquid rocket engine, the configuration of the pressure swirl injector used in liquid rocket engines should be as simple as possible. Thus the pressure swirl injector with tangential inlet ports is widely used in liquid rocket engine. Commonly, this type of pressure swirl injector consists of tangential inlet ports, a swirl chamber, a converging spin chamber, and a discharge orifice. And it can be further divided into converge-end swirl injector and open-end swirl injector based on whether there is a converging spin chamber. The atomization of the liquid propellants includes the formation of liquid film, primary breakup and secondary atomization. And the back pressure and temperature in the combustion chamber could have great influence on the atomization of the injector. What's more, when the combustion instability occurs, the pressure oscillation could further affects the atomization process. This paper reviewed the atomization mechanism and the performance of the pressure swirl injector, which include the formation of the conical liquid film, the breakup of the conical liquid film, the atomization characteristics of the conical liquid film, the effects of the rocket engine environment, and the response of the injector and atomization on the pressure oscillation.

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